

DLP6750Q1EVM

Phased Light Modulator (PLM) Evaluation Module (EVM) Kit

User's Guide – Rev 15

FFE Alpha (NIR PLM)

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EVM Overview and Assembly

EVM Overview

- The Texas Instruments (TI) DLP6750Q1EVM is an Evaluation Module for the 0.67 DLP® Products Phase Light Modulator (PLM). The EVM allows customers to evaluate the DLP® PLM technology for applications like high dynamic range (HDR), Optical Networking, Heads Up Display (HUD), Lidar, Beam Steering etc.
 - The EVM is compatible with any 0.67" PLM device.
- Holograms can be input on HDMI or DisplayPort inputs at frame rates up to 30Hz and 60Hz, respectively. A unique phase hologram can be encoded on each of the 24 RGB input bits allowing up to a 1.44kHz hologram display rate with a 60Hz input.
- The EVM has an external mirror bias supply to adjust the PLM mirror displacements (phase delays) for different wavelengths.
- The EVM provides synchronization signals that can be used to control an external power supply controlling RGB lasers.

Kit Components

Below is a list of component in the PLM EVM kit:

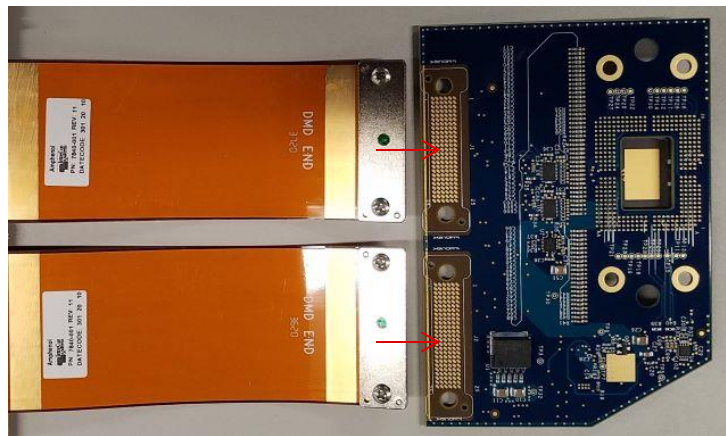
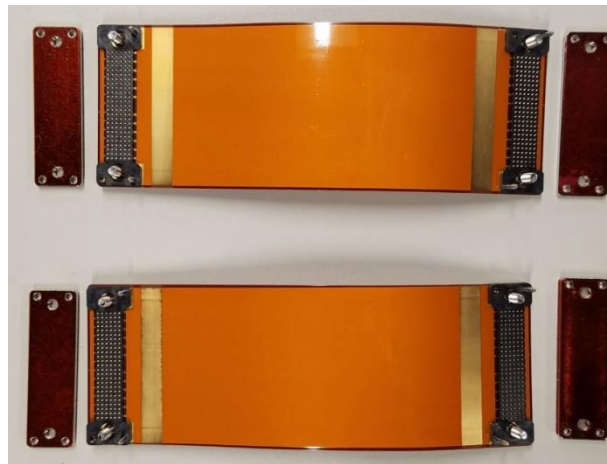
- EVM controller board (Modified version of DLPLCRC900EVM)
- PLM daughter card (Modified version of DLPLCR67EVM)
- Flex (QTY 2) (IC07840001, IC07840002, IC07840003)
- 0.67 PLM

Needed components not provided:

- USB cable
- HDMI cable (type A) or DisplayPort cable
- Electronics power supply: Recommended supply is CUI SDI65-12-U-P6, Desktop AC Adapters 60W 12V 5A, 2.5x5.5
- Mirror Bias cable: Molex 15134-0405, Cable Assemblies 4 Circuit 450MM Picoblade
- Mirror Bias power supply: Low ripple DC lab supply for PLM mirror bias: 0 to 3V, minimal current (<100mA)
- Optionally a low ripple DC lab supply for PLM electrode voltage (Voffset): 10V, minimal current (<100mA)

EVM Assembly

- With a screwdriver loosen the screws on the flex to remove the backer plates on both ends of the two flex cables.
- Insert the ends of the flexes marked DMD into the PLM board
- Hold the backer in place on the other side of the board and tighten the screws. Do not over tighten.

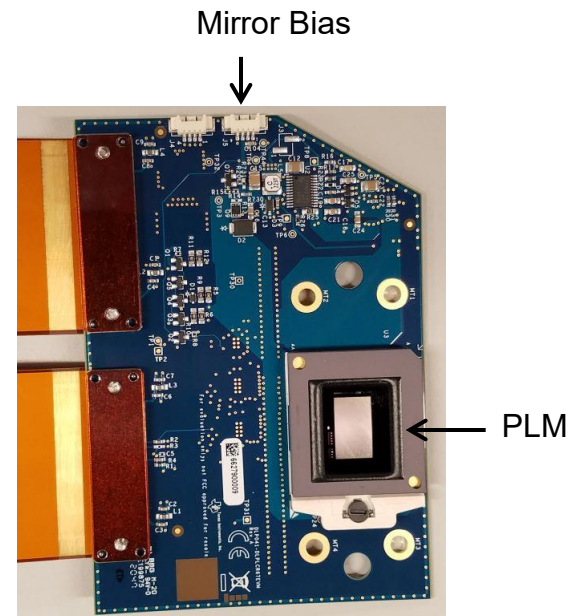
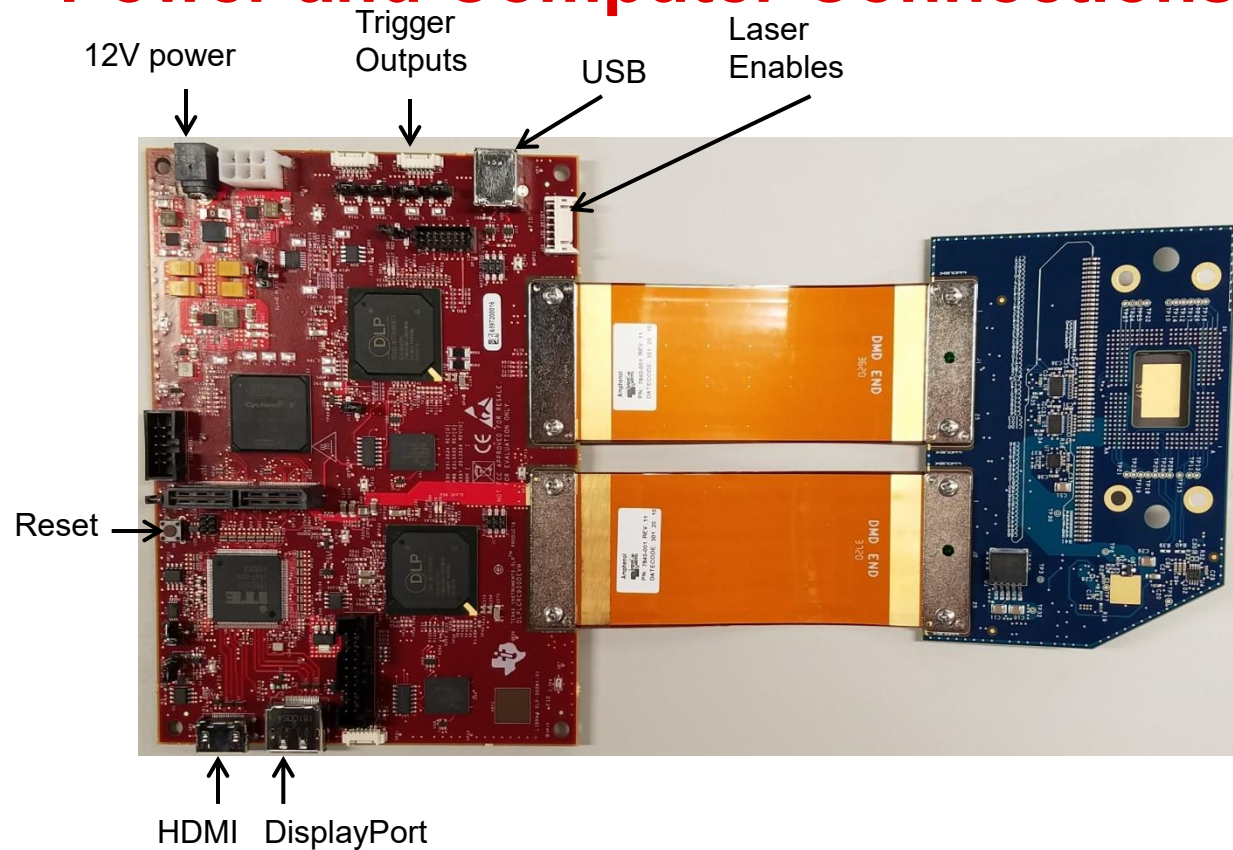


EVM Assembly

- Insert the other end of the two flexes into the controller board.
- Again, hold the backer in place on the other side of the board and tighten the two screws. Do not over tighten.
- The bottom figure shows the completed assembly



Power and Computer Connections



Mirror Bias and Electrode Voltages

- The difference between the Voffset (electrode) voltage and the Mirror Bias voltage determines the maximum displacement of the PLM mirrors.
- The EVM provides a connector on the PLM board (J5) to input the Mirror Bias voltage
- The EVM generates the Voffset (electrode) voltage on board but it can also be supplied externally.
- Mirror Bias is used to adjust the mirror displacement for different wavelength applications
 - Proposed method for mirror bias optimization is found in a subsequent page
- Specifications for Mirror Bias and Voffset are in the table below.
- Until tuned, start with the Mirror Bias at a relatively high voltage like 4V. This avoids overdriving the mirrors
- **Power Sequencing:**

- **Powering On:**

1. Mirror bias
2. EVM (12V)

- **Powering Off:**

1. EVM (12V)
2. Mirror bias

PARAMETER	MIN	NOM	MAX	UNIT
Mirror Bias Supply				
Voltage	0		5	V
Current			10	mA
Voffset (TP3)				
Voltage		10		V
Current			100	mA

PLM Board Mirror Bias and Electrode Voltage Connector (J5)

- Connector part number: Molex 532610471
- Pinout is shown below:
- Mating Cable: Molex 15134-0405 (connector can be cut off of one end to connect to power supply)
- Mating Connector and Crimp Pins: Molex 51021-0400, 50058-8000
- To use external Voffset move R730 to R731 on the PLM board

Pin	Signal	Comment
1	PLM_TP	Mirror Bias Voltage
2	GND	
3	Ext_Voffset	External Voffset (electrode Voltage)
4	GND	

DLPC900 Board Laser Enable and Trigger Connector Pinouts

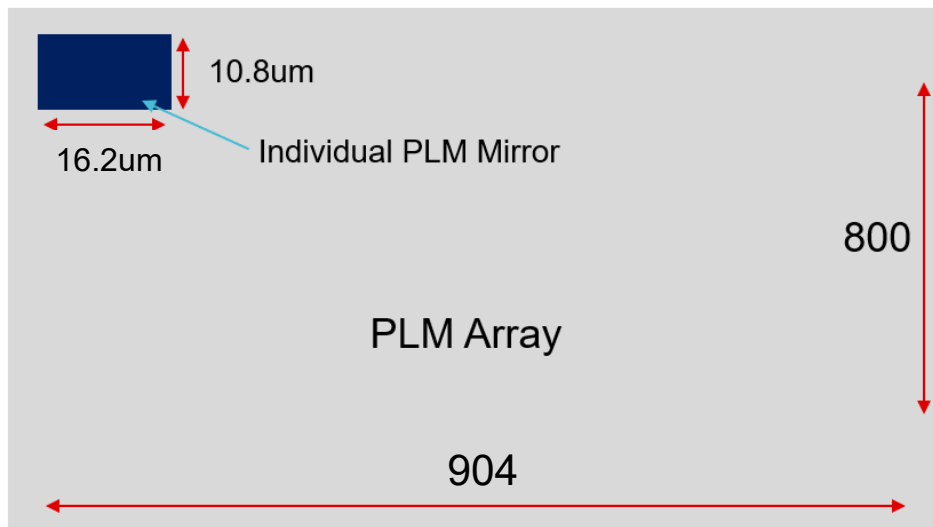
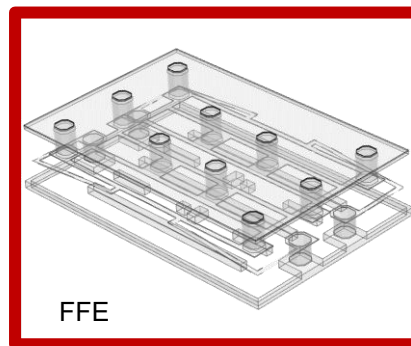
Laser Enable Connector (J26) – JST S14B-PHDSS(LF)(SN)(P)		
Pin	Signal	Comment
1	12V	
2	12V	
3	12V	
4	NC	
5	RED Enable	Asserted during Red display time. 3.3V
6	Green Enable	Asserted during Green display time. 3.3V
7	Blue Enable	Asserted during Blue display time. 3.3V
8	Reserved	Do not connect
9	Reserved	Do not connect
10	Reserved	Do not connect
11	3.3V	
12	GND	
13	GND	
14	GND	

Trigger Outputs (J27) – Molex 532610671		
Pin	Signal	Comment
1	VCC_TRIG1	J25 select 3.3V or 1.8V for TRIG1 output
2	TRIG1	
3	GND	
4	VCC_TRIG2	J28 select 3.3V or 1.8V for TRIG2 output
5	TRIG2	
6	GND	

NIR PLM Operational Properties

Device Specs

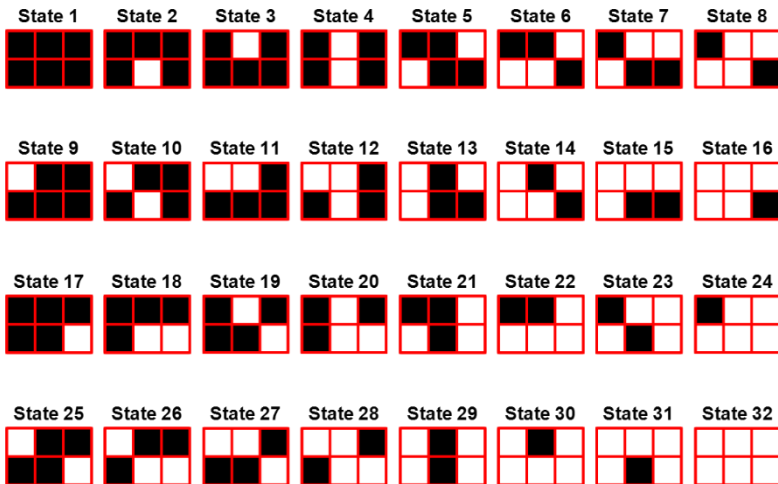
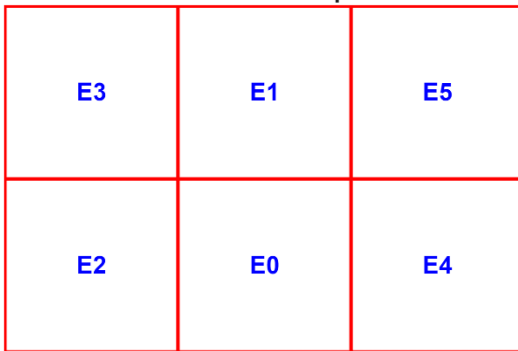
Device Spec	NIR PLM
Pitch	16.2um x 10.8um
# States	32
State Linearity	0.99
0 th order Efficiency	~80%
Wavelength Range	750nm to 1630nm
Switching Speed	~50us
Array size	0.67" (904x800)
EVM Frame Rate	1.4kHz



Formatting the hologram

- We recommend using the provided [MATLAB software](#) that helps generate and format holograms.
 - For those who prefer to format themselves, carefully consider the details on this slide.
- 6 memory cells are under each element, making the addressable memory 2712 x 1600
 - Electrodes 5 (E5) and 2 (E2) should be assigned the same value
- Since the EVM expects 2716 x 1600, you need to zero pad the left and right sides with 2 columns.
- Note the electrode map configuration and the corresponding electrode mappings by state
- The .67 EVM does a horizontal (left/right) flip, so the final encoded hologram needs to be flipped horizontally!**
 - See following slides for illustration.

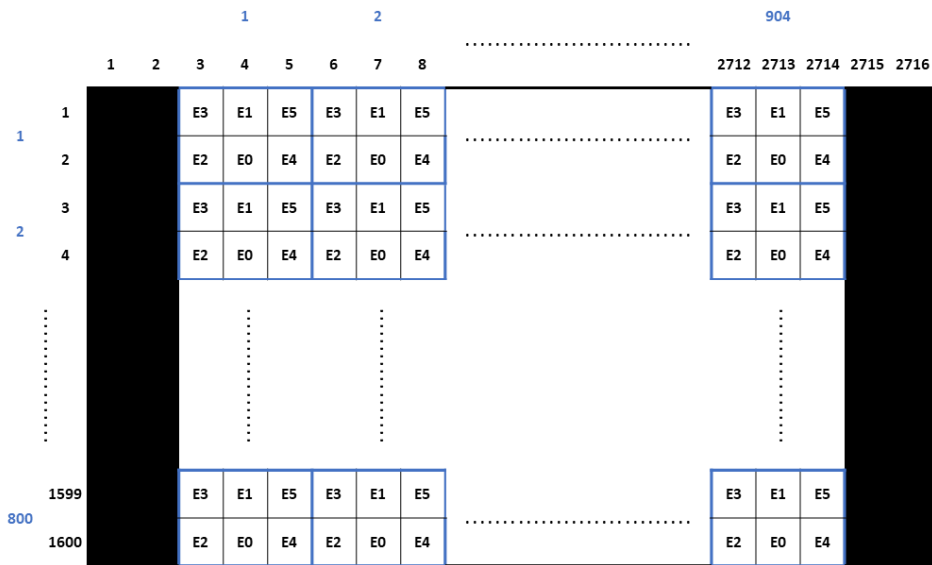
Electrode Map



Phase State	Electrodes (5,4,3,2,1,0)
1	000000
2	000001
3	000010
4	000011
5	100100
6	100101
7	100110
8	100111
9	001000
10	001001
11	001010
12	001011
13	101100
14	101101
15	101110
16	101111
17	010000
18	010001
19	010010
20	010011
21	110100
22	110101
23	110110
24	110111
25	011000
26	011001
27	011010
28	011011
29	111100
30	111101
31	111110
32	111111

NIR Formatting

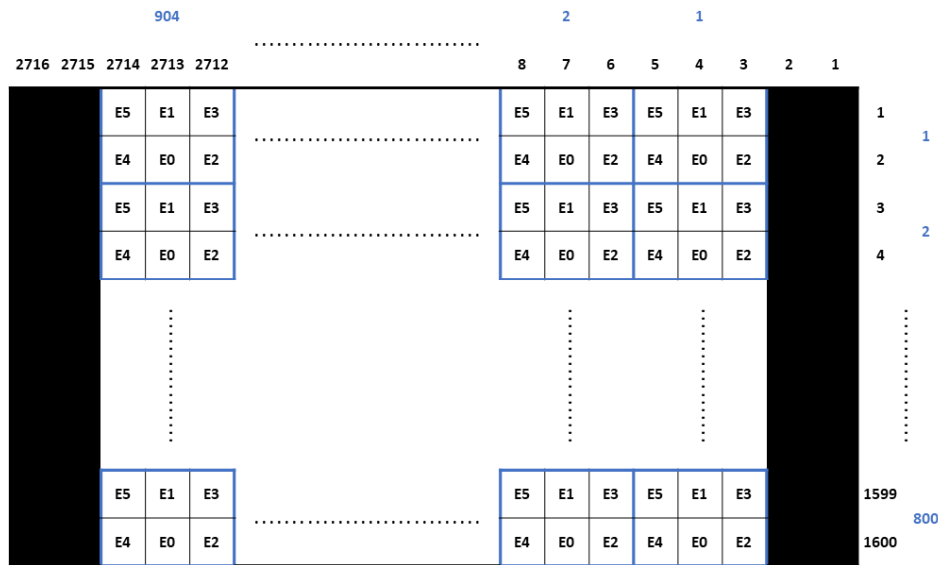
- The image below shows how the electrode configuration for the formatted (encoded) hologram array **BEFORE** the horizontal (left/right) flip has been applied.
- Note the 2 columns of zero padding on the left and right side.



Phase State	Electrodes (5,4,3,2,1,0)
1	000000
2	000001
3	000010
4	000011
5	100100
6	100101
7	100110
8	100111
9	001000
10	001001
11	001010
12	001011
13	101100
14	101101
15	101110
16	101111
17	010000
18	010001
19	010010
20	010011
21	110100
22	110101
23	110110
24	110111
25	011000
26	011001
27	011010
28	011011
29	111100
30	111101
31	111110
32	111111

NIR Formatting

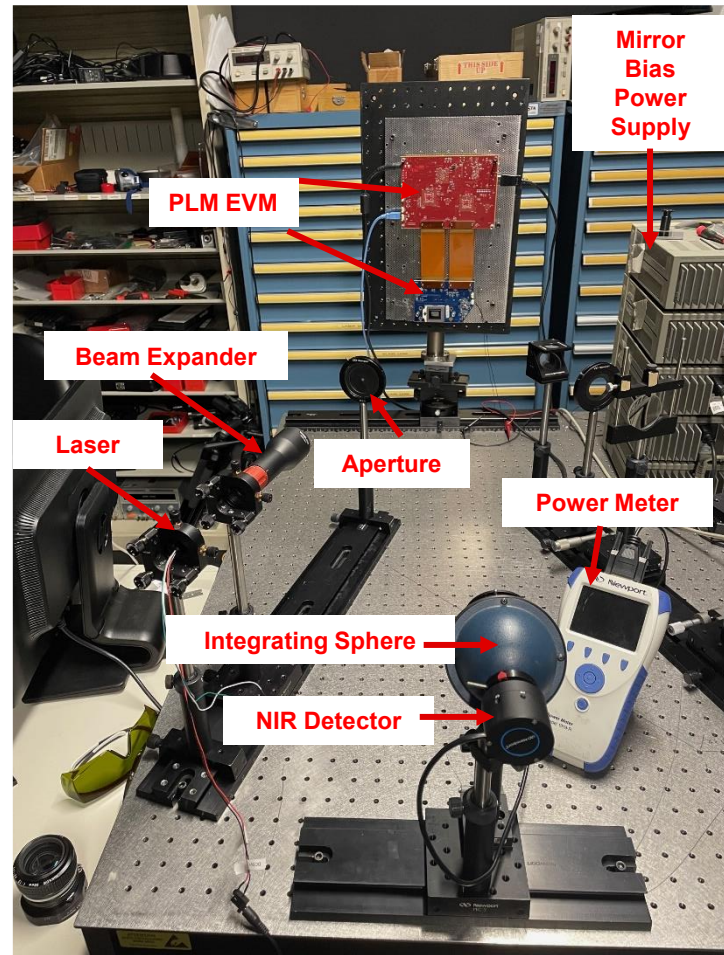
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- Note the 2 columns of zero padding on the left and right side.



Phase State	Electrodes (5,4,3,2,1,0)
1	000000
2	000001
3	000010
4	000011
5	100100
6	100101
7	100110
8	100111
9	001000
10	001001
11	001010
12	001011
13	101100
14	101101
15	101110
16	101111
17	010000
18	010001
19	010010
20	010011
21	110100
22	110101
23	110110
24	110111
25	011000
26	011001
27	011010
28	011011
29	111100
30	111101
31	111110
32	111111

NIR PLM – Tuning Mirror Bias

- As discussed previously, the mirror bias is used to tune the PLM mirror displacements / phase delay for any wavelength in its wavelength range.
- Important notes:
 - Mirror bias is specific to a device.
 - Never have the EVM powered on and mirror bias at a voltage lower than the optimal value. This can cause damage to the mirrors.
 - Do not set mirror bias below -0.5 V.
- Setup Notes:
 - We found this setup to be useful for tuning mirror bias.
 - The integrating sphere is helpful to ensure all of the laser energy off of the PLM is being collected by the detector
 - Beam expander and aperture helped with collimation and underfilling PLM array



NIR PLM – Tuning Mirror Bias

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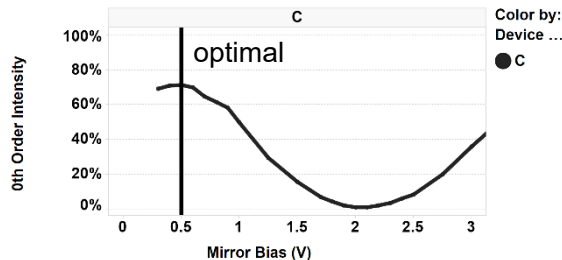
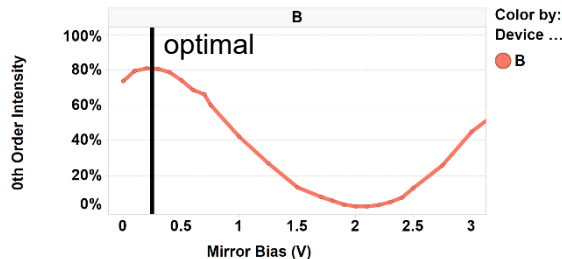
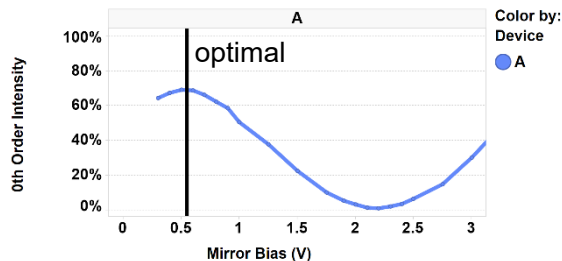
Recommended instructions:

- Set mirror bias to 3V and illuminate NIR PLM with laser
- Align zeroth order from PLM to the center of the NIR detector
- Use binary pattern (state 32 and state 1) provided.
 - If you choose to create this pattern on your own, choose state 32 for odd mirror columns and state 1 for even mirror columns.
 - At the optimal mirror bias (next step), state 32 mirrors should be displaced $\lambda/2$ relative to state 1 mirrors and provide a 2π phase delay due to reflection.
- Tune voltage on mirror bias power supply until zeroth order is maximized.
- The optimal mirror bias will be $\sim 0.1V$ higher than the voltage above. This should provide phase delays that cover the 2π range for this wavelength.



CGH_States_32_01

1550 nm Tuning Examples



PLM Mechanical

- The PLM mechanical package is the same as the DLP670S DMD. Information can be downloaded at:
 - 3D CAD Model: <https://www.ti.com/lit/zip/dlpm044>
 - Mounting and socket: <https://www.ti.com/lit/zip/dlpc128>
 - Mechanical ICD: [Mechanical ICD](#)
 - Hardware design resources: [p67 PLM HW](#)

Alpha Sample Characteristics

Alpha Sample – Overview

- Alpha samples are proof of concept designed to verify that TI can build a PLM to work in NIR space.
- The purpose of these samples is to get customers NIR PLM as quickly as possible, and further design iterations are in work to optimize the device for performance
- These units and future NIR samples are compatible with the visible PLM EVM kit ([DLP6750Q1EVM](#)) and the EVM instructions should be followed.
 - [.67 Users Guide](#).
 - Alternatively, the device can be run using the two kits below:
 - EVM controller board: <https://www.ti.com/tool/DLPLCRC900DEV>
 - Daughter card: <https://www.ti.com/tool/DLPLCR67EVM>
- Operation up to 1630nm has been verified with this pixel
- 32 phase state accessibility achieved
- Measured values show general agreement to model results. Tests still ongoing...

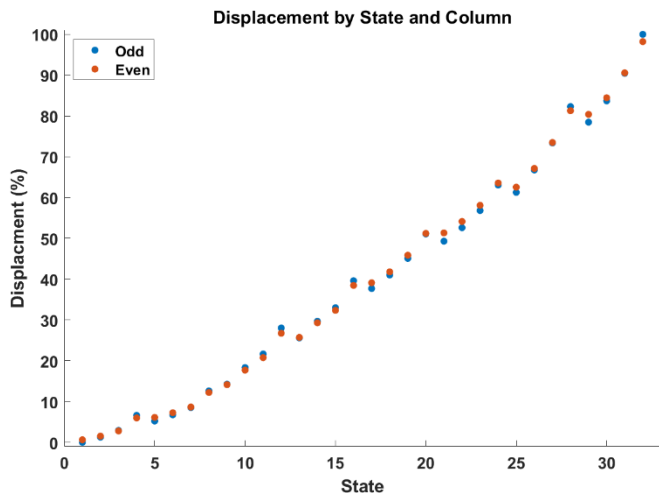
Alpha Sample – Characteristics

- Our Alpha samples have 3 primary characteristics that we are actively working to fix. Solutions are known and will be implemented in subsequent samples: (1) Random, unpowered tilted mirrors, (2) displacement variation by column, and (3) tilt variation by column and state.
- Details in the following slides.

Sample	Unpowered Tilted Mirrors	Displacement Variation by Column	Tilt Variation by Column and State	Approximate Timeline
Alpha	High	High	High	Now
Alpha'	None	High	High	4Q25
Beta	None	Moderate	Moderate	2Q26
Gamma	None	None	Low	3Q26
Delta	None	None	Extremely low	4Q26

Alpha Sample – Displacement Variation by Column

- Odd mirror columns versus even mirrors columns exhibit slightly different displacements and thus phase delays.
- This can accounted for when states are discretized during hologram generation.
- This may result in a ‘binary grating’ artifact in your test setup.

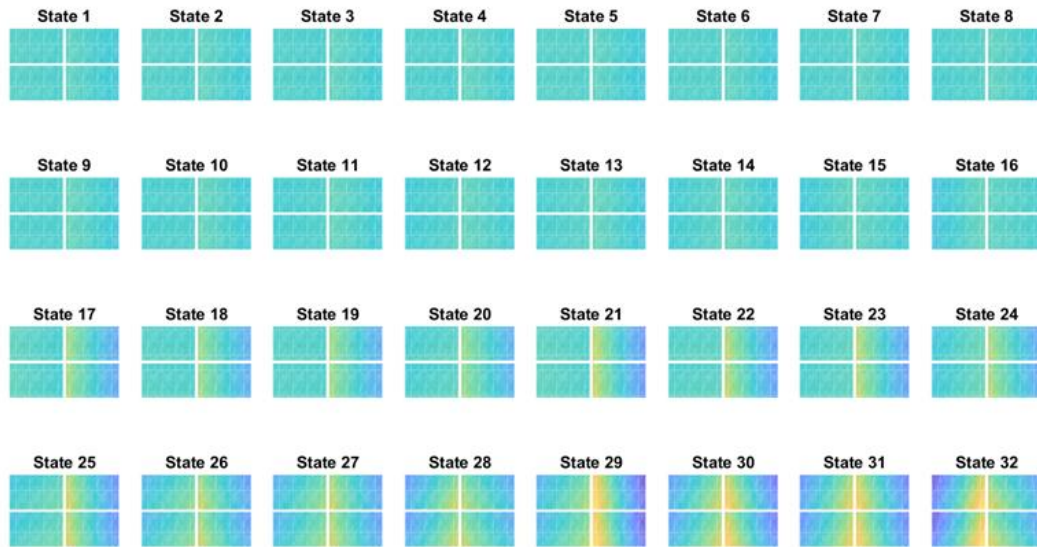
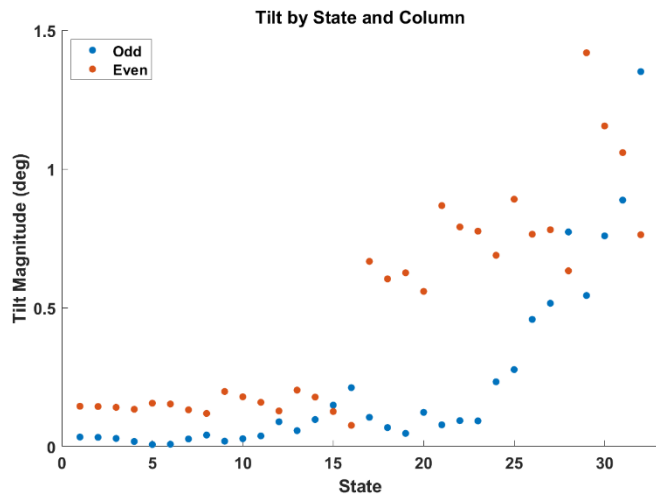


Phase State	Typical Odd Displacement	Typical Even Displacement
1	0.00%	0.64%
2	1.27%	1.53%
3	2.93%	2.80%
4	6.62%	5.99%
5	5.22%	6.11%
6	6.75%	7.26%
7	8.54%	8.66%
8	12.61%	12.23%
9	14.27%	14.14%
10	18.34%	17.71%
11	21.66%	20.76%
12	28.03%	26.75%
13	25.61%	25.73%
14	29.68%	29.30%
15	32.99%	32.36%
16	39.62%	38.47%

Phase State	Typical Odd Displacement	Typical Even Displacement
17	37.71%	39.11%
18	41.02%	41.78%
19	45.10%	45.86%
20	51.08%	51.21%
21	49.30%	51.34%
22	52.61%	54.14%
23	56.82%	58.09%
24	63.06%	63.57%
25	61.27%	62.55%
26	66.75%	67.13%
27	73.38%	73.50%
28	82.29%	81.27%
29	78.47%	80.38%
30	83.69%	84.46%
31	90.45%	90.57%
32	100.00%	98.22%

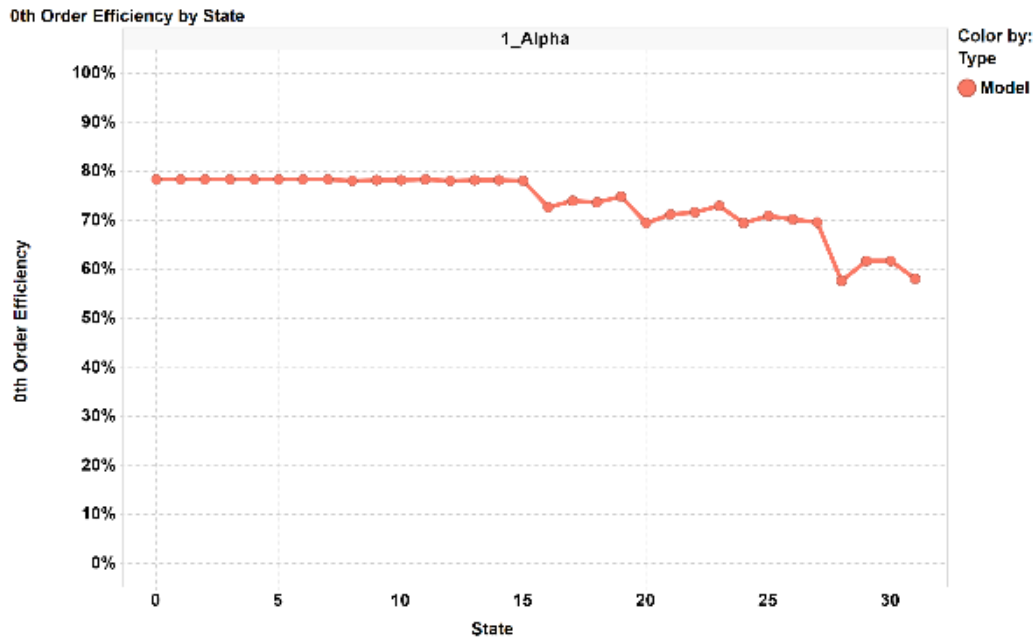
Alpha Sample – Tilt Variation by Column and State

- Graph below shows modeled tilts for displacement tuned for ~1600 nm
- This may result in a 'binary grating' artifact in your test setup



Global State 0th Order Efficiency

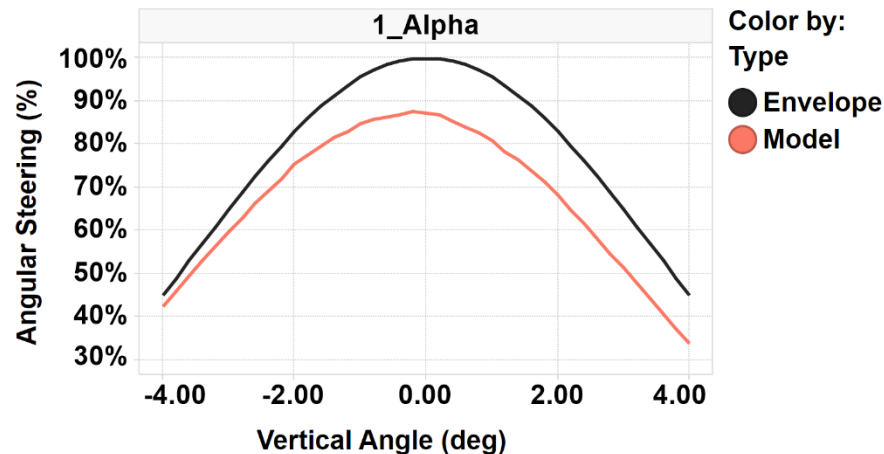
- 0th order efficiency is reduced largely due to mirror tilt effects at higher states.
- Modeled results below are for displacements tuned for 1550 nm.



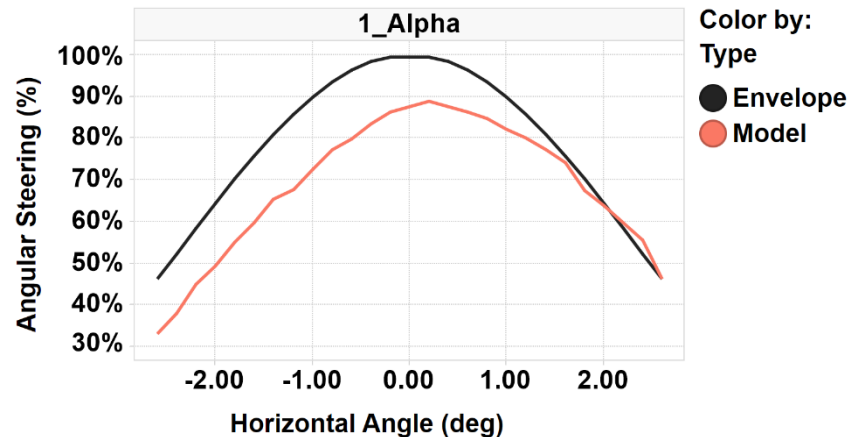
Angular Steering Efficiency – Horizontal and Vertical

- Graphs for angular steering efficiency in vertical and horizontal directions.
- Envelope is included to show the theoretical max.
- The modeled curves are not centered at 0 degrees due to a bias tilt that shifts the actual envelope.

Vertical - 1550nm

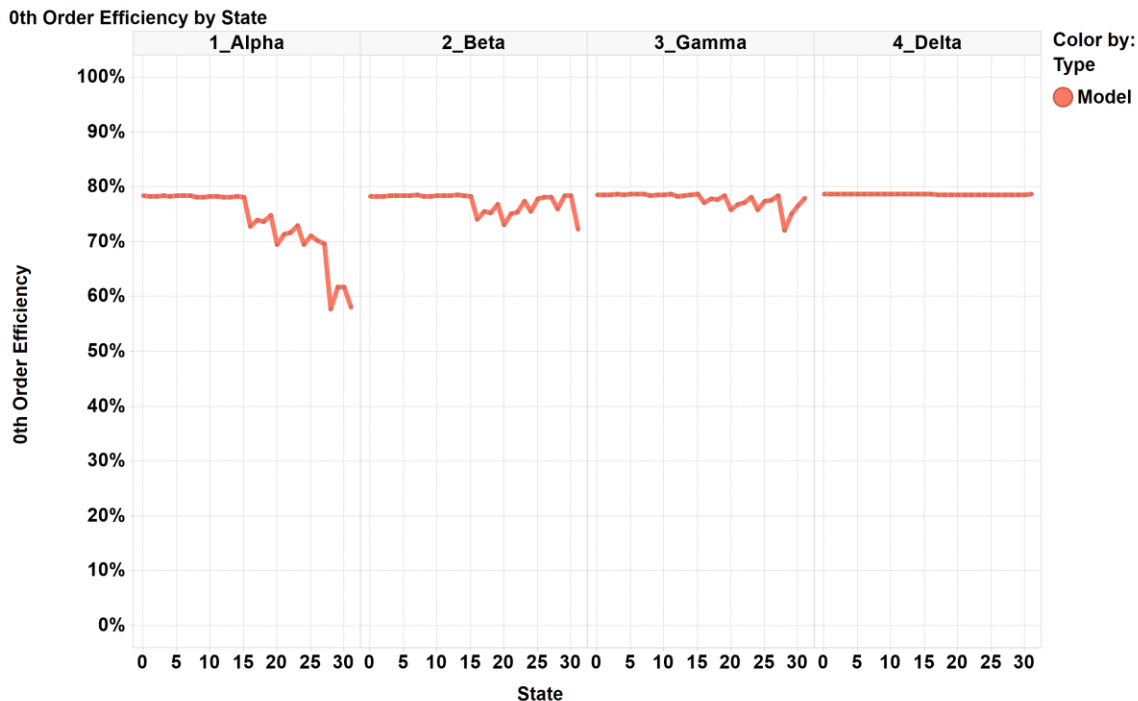


Horizontal - 1550nm



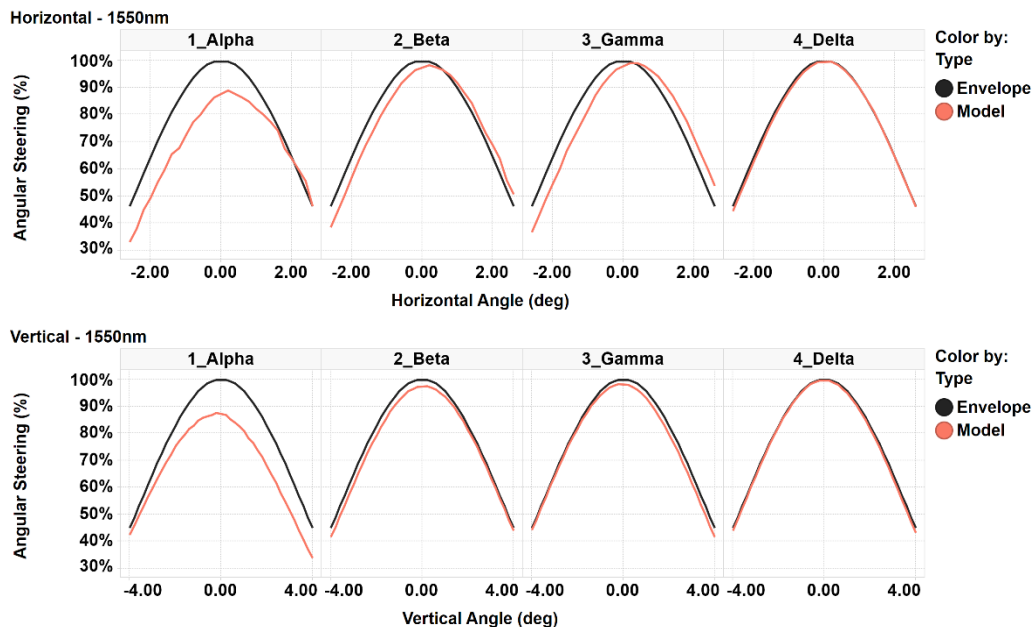
Alpha Sample – Initial Modeling and Future Builds

- Unpowered 0th order efficiencies perform reasonably well at 1550 nm.
 - Measured values are meeting and in some cases exceeding our initial modeled results.
- Samples enable phase modulation and can be used for prototyping to gain familiarity with NIR PLM
- 0th order efficiencies by state look reasonable at 1550nm for most states
- **Our future builds, as shown on the right, will address and eliminate variation in 0th order efficiency by state**



Alpha Sample – Initial Modeling and Future Builds

- Angular steering efficiency shows signatures of biased tilt and is about 10-15% less than the theoretical max
- Our future builds, as shown on the right, will address the deficiencies in angular steering efficiency.**



Summary

- We are very excited for our customers to start development with the NIR PLM alpha samples!
- These samples are from our first build and we see room for improvement and know the paths to get there.
- We hope these provide solutions to your needs / applications.
- **Please give us feedback on these units while we work to provide you with updated samples in the near future!**

Device Spec	NIR PLM	Customer Feedback
Pitch	16.2um x 10.8um	
# States	32	
State Linearity	0.99	
0 th order Efficiency	~80%	
Wavelength Range	750nm to 1630nm	
Switching Speed	~50us	
Array size	0.67" (904x800)	
EVM Frame Rate	1.4kHz	
Other Feedback		

Getting Started with LightCrafter

EVM Instructions - Connecting to Computer and applying power

- Connect the mirror bias supply to J5 on the EVM PLM board
- Turn on the mirror bias supply first and then turn on 12V power supply
- Connect 12V power to the assembled EVM power connector
- Adjust mirror bias supply to the voltage for desired displacement
- Connect USB cable from computer to EVM's USB connector
- Connect either HDMI or DisplayPort cable from computer or video source to the EVM input
- Configure the computer or video source to output for 2716x1600
 - HDMI is limited to 2716x1600 @ 30Hz
 - DisplayPort can support 2716x1600 @ 60Hz

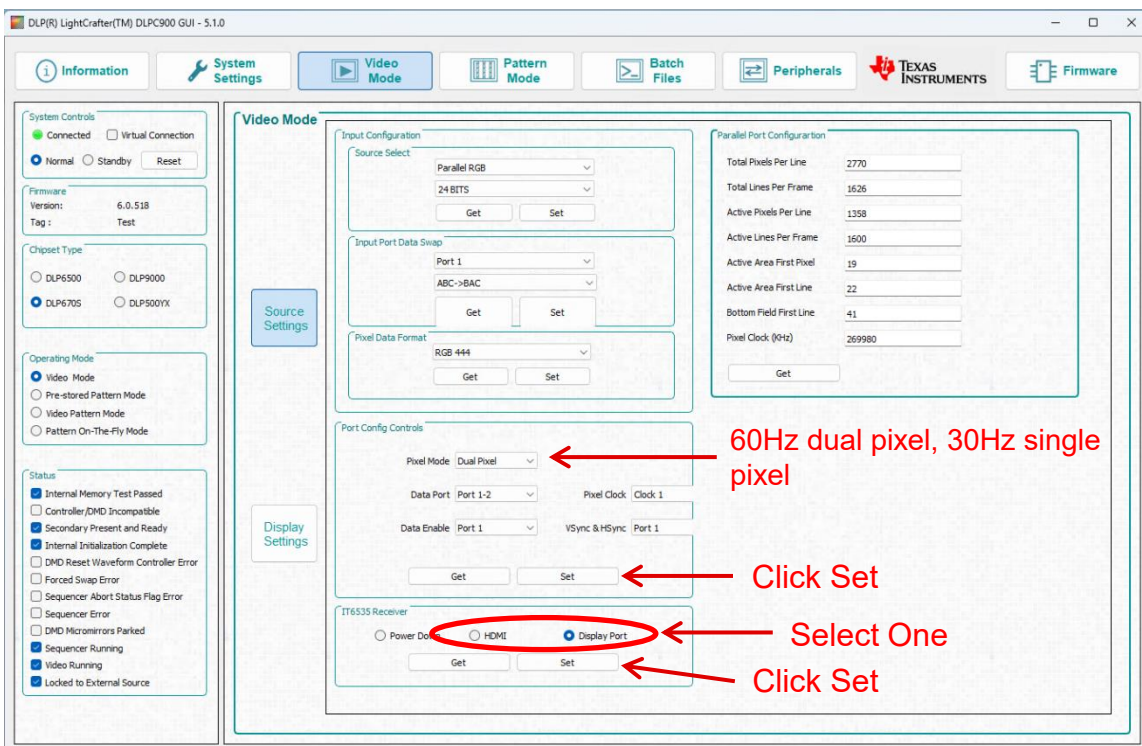
EVM Instructions – GUI software

- Install GUI software [here](#)
 - Version 5.1 is required
- Start GUI software
- GUI should automatically connect to the EVM and the connected box will be green



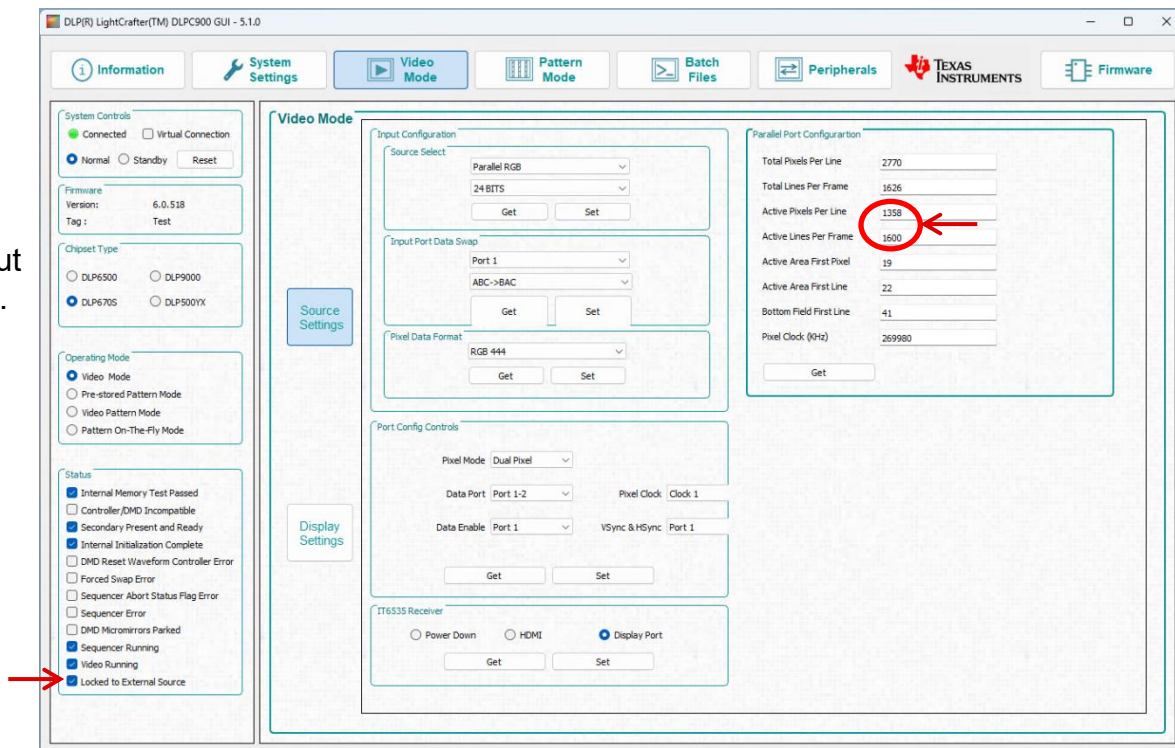
EVM Instructions – Video Mode Settings

- In the Video Mode window, verify the Input Configuration settings are the same as the image provided.
- In the Port Config Controls, select and “Set” the appropriate pixel mode based on the following:
 - Dual pixel for DisplayPort (60Hz)
 - Single pixel for HDMI (30Hz)
- In the IT6535 Receiver, select and “Set” the appropriate receiver.



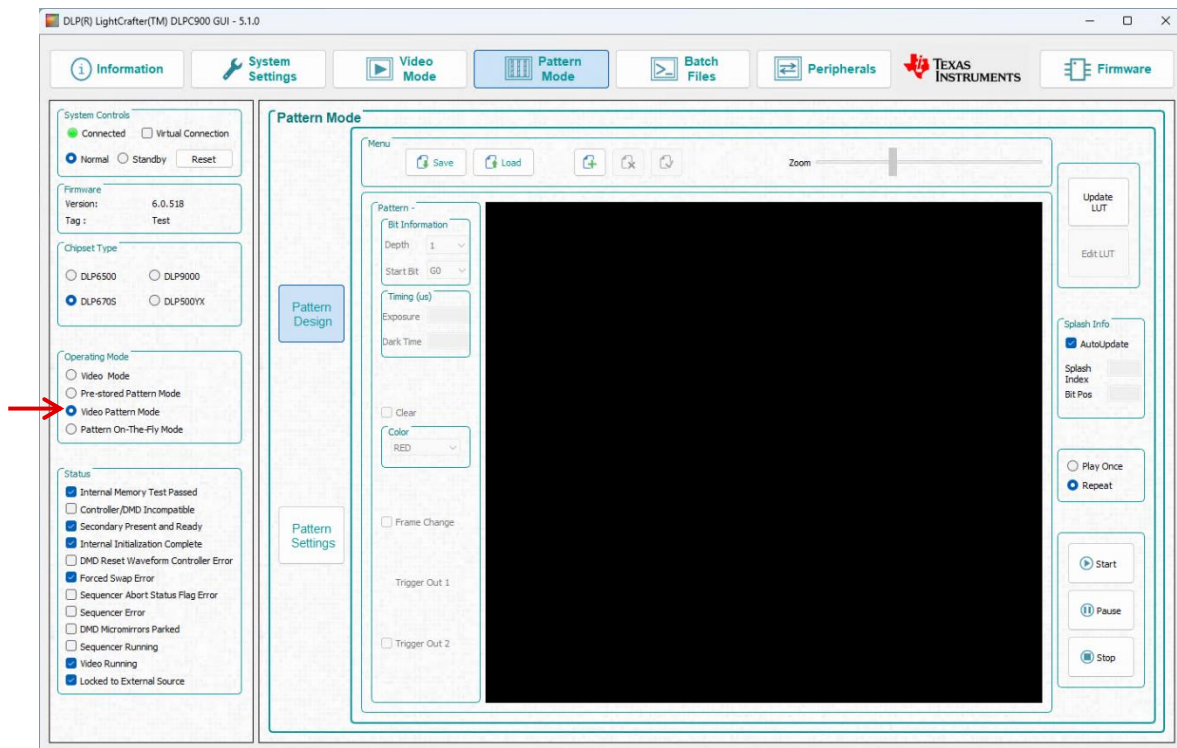
EVM Instructions – Video Mode Settings (60Hz example)

- Check that the Locked to External Source is checked (will take a few seconds)
- Click on the Get button in the Parallel Configuration box and verify that the active area is correct. Note that the Active pixels per line will be half the input as each ASIC processes half the image.
 - For example 2716x1600 should show 1358 active pixels and 1600 active lines.



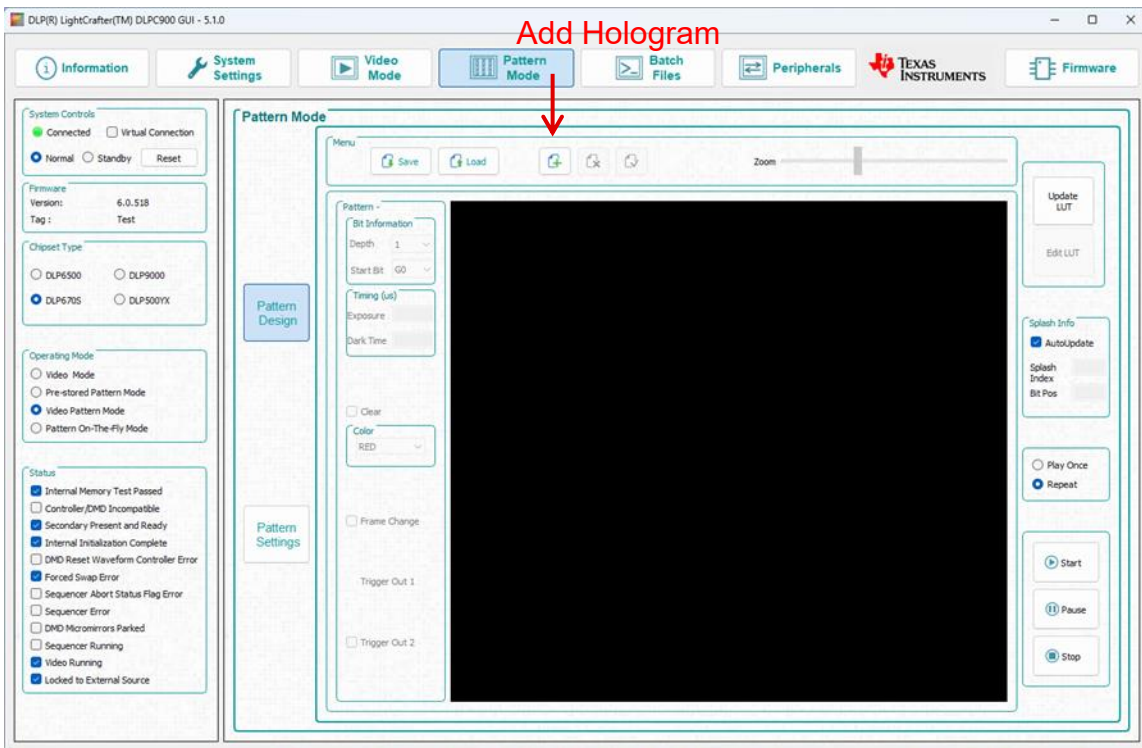
EVM Instructions – Video Pattern Mode Settings (single hologram example)

- Click on Video Pattern Mode



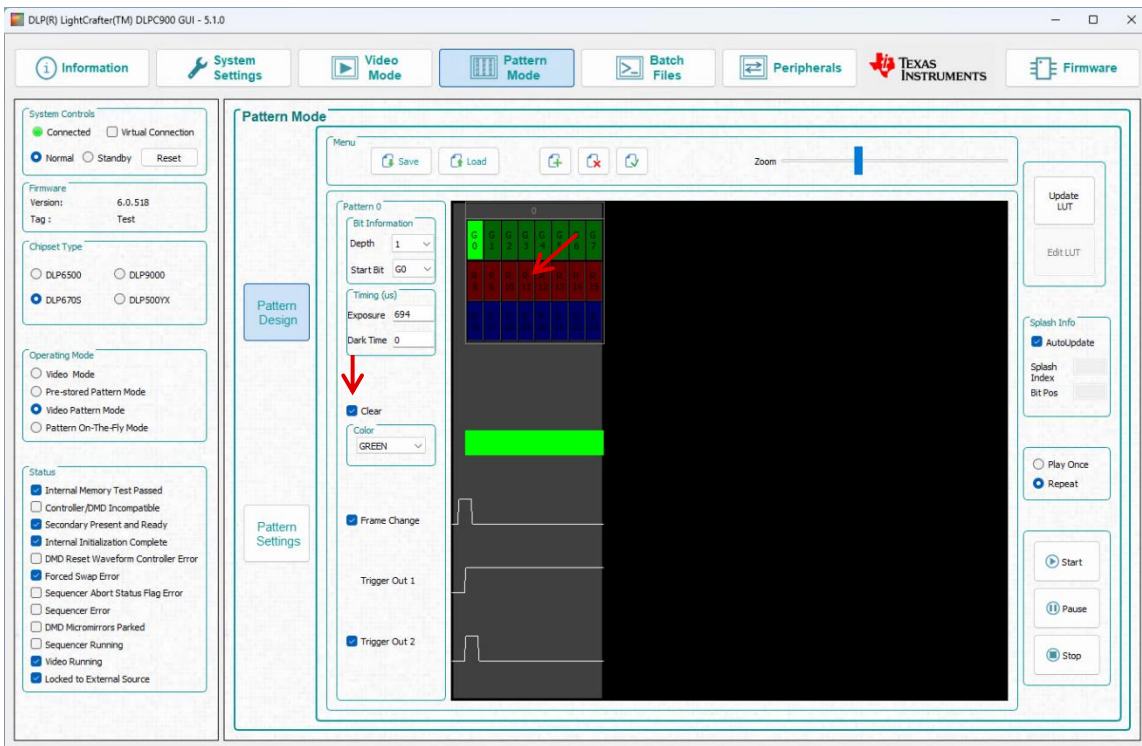
EVM Instructions – Video Pattern Mode Settings (single hologram example)

- Click on Video Pattern Mode
- Click on Add Hologram button



EVM Instructions – Video Pattern Mode Settings (single hologram example)

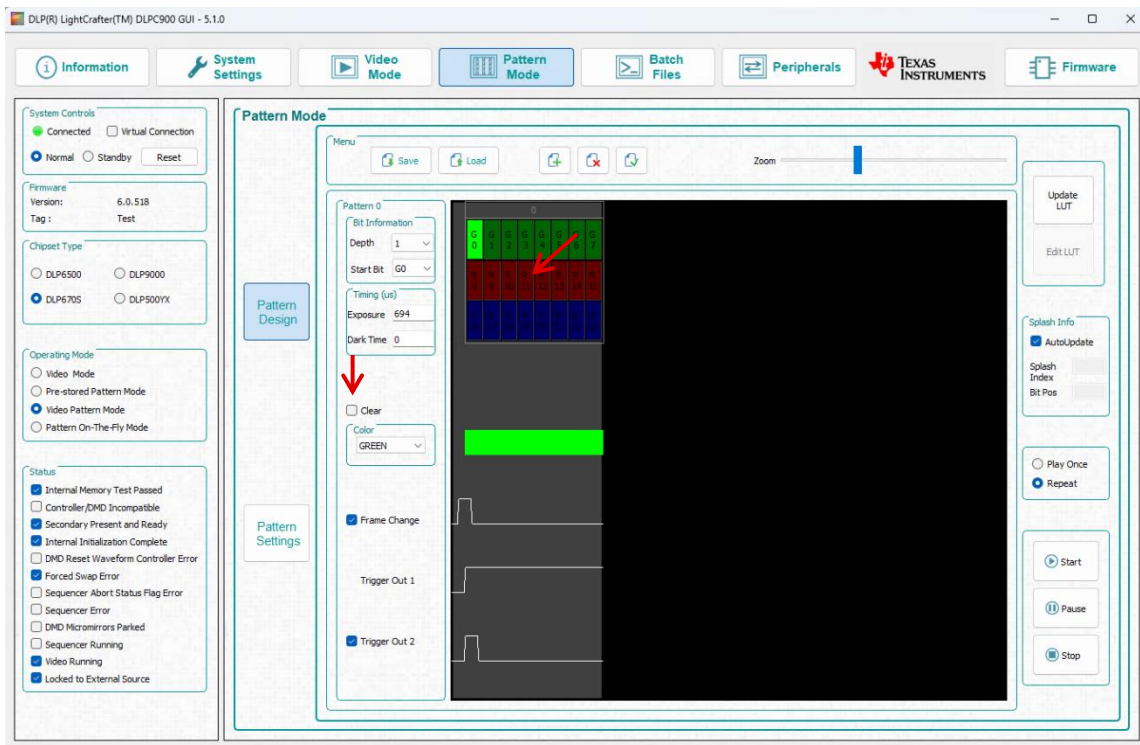
- Click on Video Pattern Mode
- Click on Add Hologram button
- Click in the frame to select it
- Uncheck the Clear button



EVM Instructions – Video Pattern Mode Settings (single hologram example)

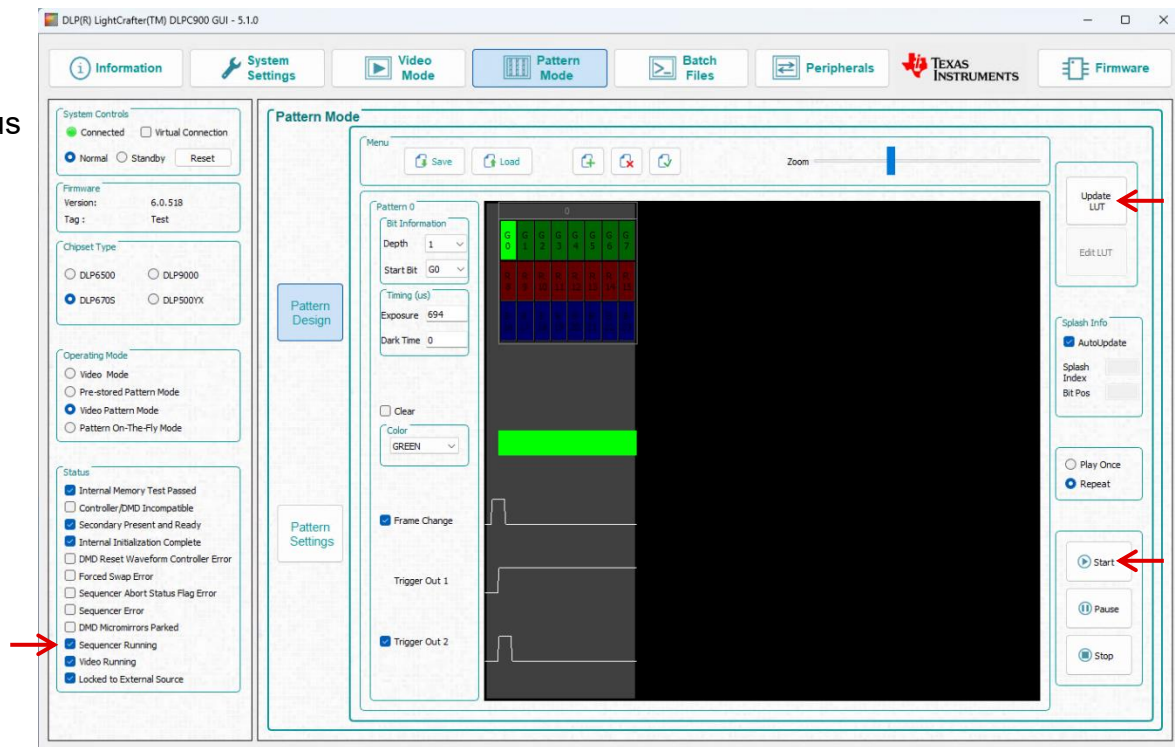
- Click on Video Pattern Mode
- Click on Add Hologram button
- Click in the frame to select it
- Uncheck the Clear button

The pattern created here displays a single hologram. Since the Clear button is unchecked, the single hologram continues to be displayed after the 694us subframe until the end of the 60Hz frame (~16.7ms).



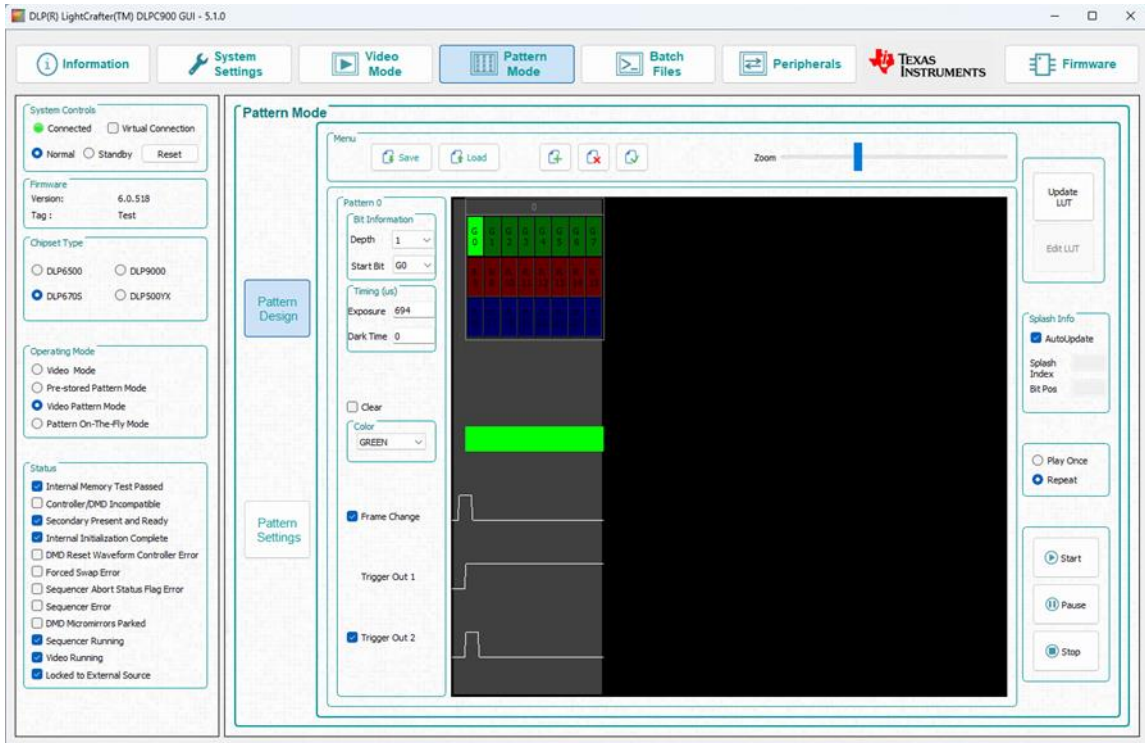
EVM Instructions – Video Pattern Mode Settings (single hologram example)

- Click on Update LUT
- Click on Start
- The Sequence Running box in the Status window should become checked.
- To change parameters click on the Stop button, change parameters, click on Update LUT and then click on Start



EVM Instructions – Video Pattern Mode Settings

- More complex patterns can be created, saved, and later loaded. Please see the supplemental section to see how to create them.
- The GUI allows a pattern of different holograms that are input on the 24 RGB input bits to be displayed within the frame time.



EVM Instructions – Source Setup

- In computer display settings, configure the monitor to the EVM to be the same resolution as the formatted hologram (2716x1600).
- Ensure the scale in display settings is set to 100%
- You can use a picture viewer software such as Windows Photo Viewer to display the formatted hologram full screen on the display. Make sure the picture viewer software does not put a boarder up around the outside.
- To view the hologram reflect a laser off the array. A laser pointer can be used but the resulting image will be noisy as only part of the array is illuminated.

Debugging EVM and Supplemental

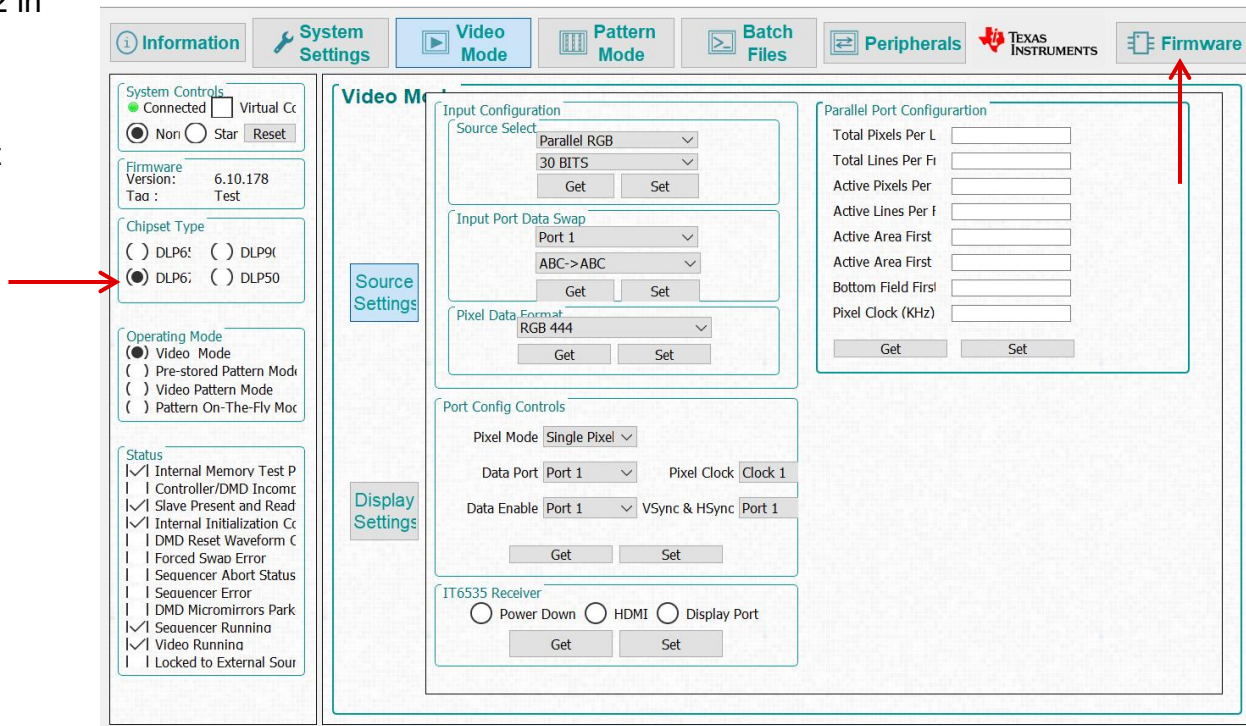
Firmware Update

Firmware Update

- The EVM will be shipped with Phased Loading firmware that has a number of supported base sequences (694us, 925us, 940us, 1400us, 1810us, 16700us, etc.).
- Firmware with a different base sequence time, or with global loading can be loaded with these instructions.

EVM Instructions – Firmware Update

- With power off, install Jumper on J12 in middle of board.
- Turn on Mirror bias followed by 12V
- Click on the DLP670S in the Chipset Type Window
- Click on the Firmware button



EVM Instructions – Firmware Update

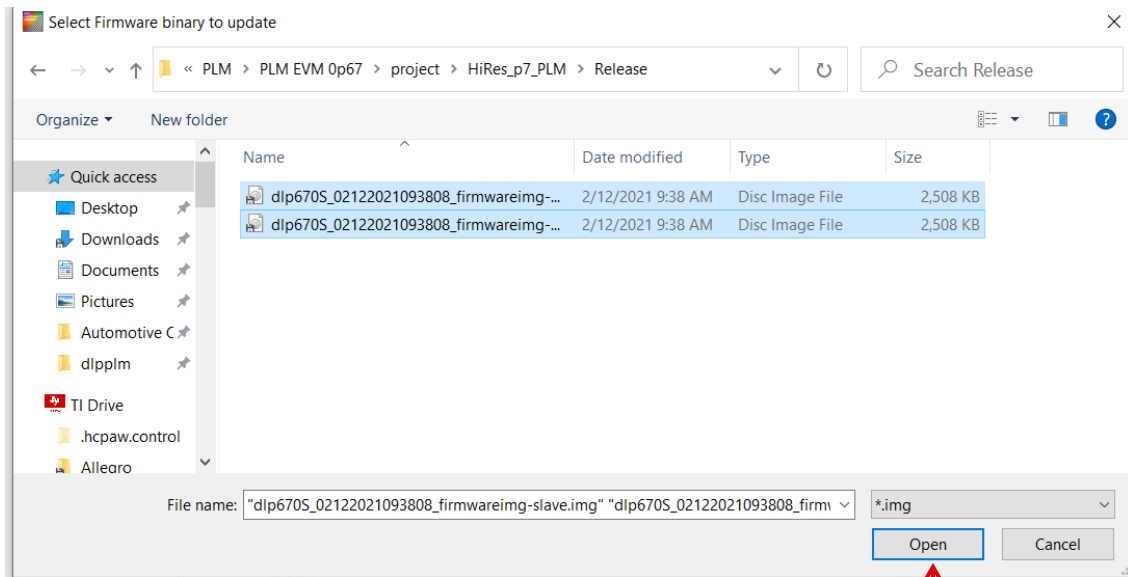
- Click on the Browse button and select both the master and slave image files

The screenshot displays the Texas Instruments EVM Firmware Update Utility interface. The 'Firmware' tab is selected in the top navigation bar. The main content area is divided into several sections:

- System Controls:** Includes 'Connected' status, 'Virtual Cor' checkbox, and 'Norm'/'Stanc' radio buttons with a 'Reset' button.
- Firmware Version:** Shows '6.10.178' and 'Test'.
- Chipset Type:** Includes radio buttons for 'DLP65C', 'DLP90C', 'DLP67C', and 'DLP500'.
- Operating Mode:** Includes radio buttons for 'Video Mode', 'Pre-stored Pattern Mode', 'Video Pattern Mode', and 'Pattern On-The-Fly Mode'.
- Status:** A list of checkboxes indicating system status, such as 'Internal Memory Test Pa', 'Controller/DMD Incompa', 'Slave Present and Ready', 'Internal Initialization Cor', 'DMD Reset Waveform Cc', 'Forced Swap Error', 'Sequencer Abort Status F', 'Sequencer Error', 'DMD Micromirrors Parker', 'Sequencer Running', 'Video Running', and 'Locked to External Sourc'.
- Firmware Section:** Contains a 'Firmware File' input field, a 'Browse' button (highlighted with a red arrow), and an 'Upload' button. Below this is a 'Status' section with a progress bar showing '0%'. There is also a 'Batch Files Stored in Firmware' section with a 'Save to File' button and a 'Delete' button.
- Power on Defaults:** Includes checkboxes for 'Start up Stat', 'Default Batd', 'Pattern Sequence Cache Size', and 'Update Flash Wait Cycle'. It also has input fields for 'Read Delay', 'Write Delay', 'Write Pulse Wid', and 'CS/Data Hold ti'.
- DMD Bus Swap:** Includes checkboxes for 'DMD Bit Swap Channel A', 'DMD Bit Swap Channel B', and 'DMD Swap AB Channels'.
- Spread Spectrum:** Includes a checkbox for 'Spread Spectrum'.
- Batch File and Patterns:** Includes checkboxes for 'Clear All Existing Pattern Images and Batch Files', 'Add Batch File label', 'Add Pattern Images (from Pattern Mode tab)', and 'Set as Default Batch File'.
- Update Firmware:** A button at the bottom right of the interface.

EVM Instructions – Firmware Update

- Select both the master and slave image files and click on the Open button



EVM Instructions – Firmware Update

- Click on the Upload Button
- GUI will erase, program and verify Master and then the slave.
- It will then cache the firmware from the Master and Slave.
- If the GUI reports a checksum mismatch click on the Upload button again to repeat the programming.
- Once the programming is complete power off 12V and remove jumper on J12

The screenshot shows the 'Firmware' tab in the Texas Instruments EVM GUI. The 'Firmware File' field is set to 'C:/Users/a0691538/TI Drive/auto/PLM/PLM EVM Op67/'. The 'Status' field is empty. The 'Batch Files Stored in Firmware' section is empty. The 'Refer to the datasheet or programmers guide for details of th' section contains various settings like 'Power up Defaults', 'Start up Stat', 'HDMI/DP Re', 'Default Batcl', 'I2C Slave Ac', 'Pattern Sequence Cache Size', 'Update Flash Wait Cycle', 'Read Delay', 'Write Delay', 'Write Pulse Wid', 'CS/Data Hold ti', 'DMD Bus Swap', 'DMD Bit Swap Channel A', 'DMD Bit Swap Channel B', 'DMD Swap AB Channels', 'Spread Spectrum', 'Batch File and Patterns', 'Clear All Existing Pattern Images and Batch Files', 'Add Batch File label', 'Add Pattern Images (from Pattern Mode tab)', 'Set as Default Batch File', and an 'Update Firmware' button. A red arrow points to the 'Upload' button.

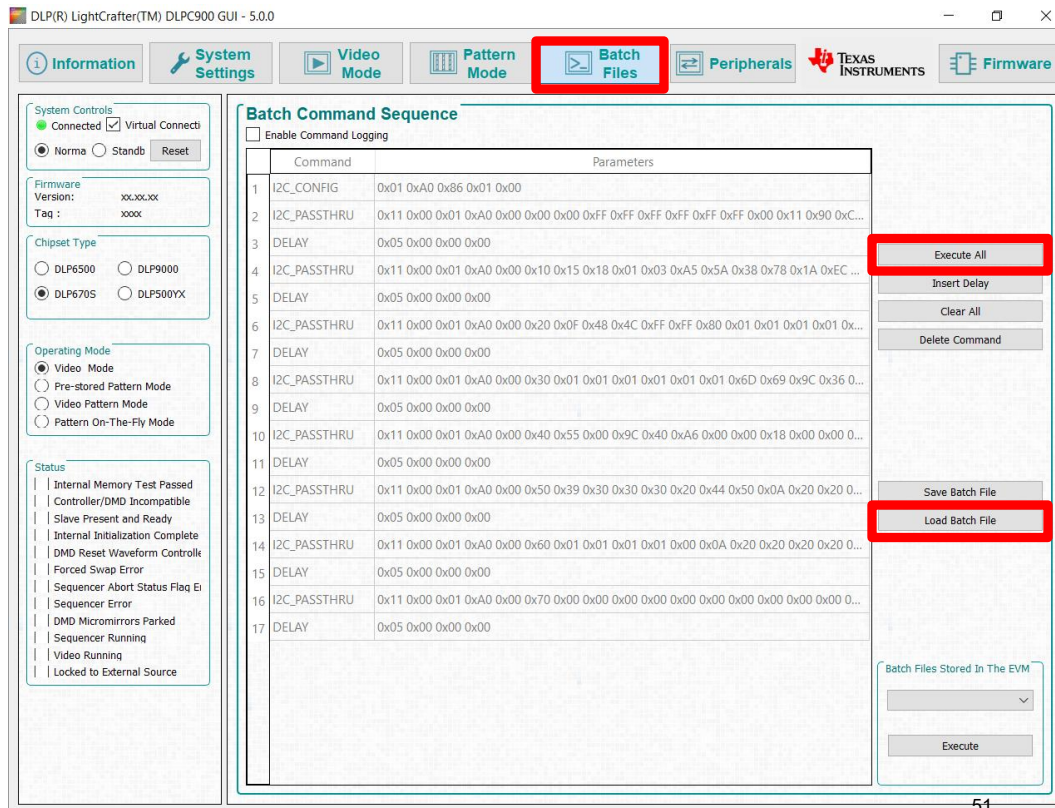
EDID Update

EVM Instructions – EDID Update

- EDIDs are provided for the HDMI or DisplayPort interface that support only 2716x1600, only 2560x1600 or both resolutions.
- The EDIDs with only one resolution are provided in case the users has difficulty getting their computer to select the desired resolution.
- We recommend using the 2716x1600 EDID as this will utilize the entire array of mirrors on the PLM.
- Using the 2560x1600 leaves 39 columns of mirrors on both sides of the array that are not utilized.

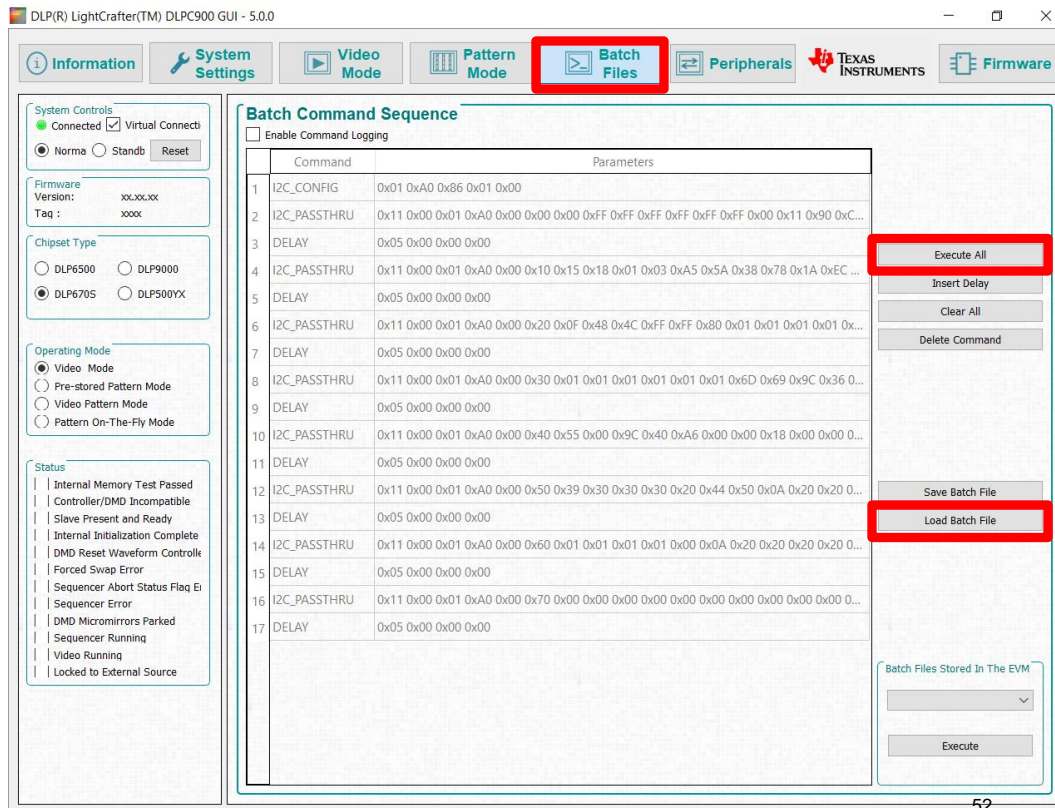
EVM Instructions – HDMI EDID Update

- Reprogramming the HDMI EDID
 - Power off the EVM
 - Install jumper on J3
 - Connect HDMI cable from computer to EVM
 - Connect USB cable from computer to EVM if not already connected
 - Power on the EVM (turn on mirror bias first)
 - Start the GUI software if not already running
 - Click on the “Batch Files” tab
 - Click on “Load Batch File”
 - Select the desired EDID batch file
 - Click on “Execute All”
 - Wait 5 seconds (note the GUI does not provide any status on the batch file execution)
 - Power off the EVM
 - Remove the jumper from J3



EVM Instructions – DisplayPort EDID Update

- Reprogramming the DisplayPort EDID
 - Power off the EVM
 - Install jumper on J4
 - Connect DisplayPort cable from computer to EVM
 - Connect USB cable from computer to EVM if not already connected
 - Power on the EVM (turn on mirror bias first)
 - Start the GUI software if not already running
 - Click on the “Batch Files” tab
 - Click on “Load Batch File”
 - Select the desired EDID batch file
 - Click on “Execute All”
 - Wait 5 seconds (note the GUI does not provide any status on the batch file execution)
 - Power off the EVM
 - Remove the jumper from J4



Trouble Shooting

- Make sure the resolution of the computer or laptop matches the holograms resolution and the resolution output to PLM (2716x1600). Set the computer to duplicate mode. This allow seeing the phase info on the computer screen and full screen mode can be selected.
- When you display the phase image on the computer, make sure it is full screen without any borders around it. The Photos program works when expanded to full screen.
- Make sure the clear checkbox is not checked.

Example Input Video Timing

Example video timings

2560x1600@30Hz.prg - Timing

H

Input Mode ☐ usec ☒ dot

DotClock 134.250 MHz

Repetition 1

Frequency 49.36 KHz

Period ☐ 20.26 us ☐ 2720 dot

Disp ☐ 19.07 us ☐ 2560 dot

Sync 0.24 us 32 dot

Backp 0.60 us 80 dot

HD Start 0.00 us 0 dot

HD Width 0.00 us 0 dot

Frontp 0.36 us 48 dot

Blanking 1.19 us 160 dot

V

Input Mode ☐ msec ☒ H

Scan Progressive

Field Setting ☒ Field-1 & 2 ☐ Field-1(=2)

TV Mode Other

Frequency 29.97 Hz

Total ☐ 33.369 ms ☐ 1647 H

Disp ☐ 32.417 ms ☐ 1600 H

Sync 0.243 ms 12.0 H

Backp 0.567 ms 28 H

Serration 1H

EQP Off

EQP fp 0.000 ms 0.0 H

EQP bp 0.000 ms 0.0 H

VD Start 0.000 ms 0.0 H

VD Line 0.000 ms 0.0 H

Frontp 0.142 ms 7.0 H

Blanking 0.952 ms 47.0 H

2560x1600@60Hz.prg - Timing

H

Input Mode ☐ usec ☒ dot

DotClock 268.500 MHz

Repetition 1

Frequency 98.71 KHz

Period ☐ 10.13 us ☐ 2720 dot

Disp ☐ 9.53 us ☐ 2560 dot

Sync 0.12 us 32 dot

Backp 0.30 us 80 dot

HD Start 0.00 us 0 dot

HD Width 0.00 us 0 dot

Frontp 0.18 us 48 dot

Blanking 0.60 us 160 dot

V

Input Mode ☐ msec ☒ H

Scan Progressive

Field Setting ☒ Field-1 & 2 ☐ Field-1(=2)

TV Mode Other

Frequency 59.97 Hz

Total ☐ 16.675 ms ☐ 1646 H

Disp ☐ 16.209 ms ☐ 1600 H

Sync 0.061 ms 6.0 H

Backp 0.375 ms 37 H

Serration 1H

EQP Off

EQP fp 0.000 ms 0.0 H

EQP bp 0.000 ms 0.0 H

VD Start 0.000 ms 0.0 H

VD Line 0.000 ms 0.0 H

Frontp 0.030 ms 3.0 H

Blanking 0.466 ms 46.0 H

Example video timings

2716x1600@30Hz.prg - Timing

H

Input Mode ☐ usec ☒ dot

DotClock 139.800 MHz

Repetition 1

Frequency 48.61 KHz

Period ☐ 20.57 us ☐ 2876 dot

Disp ☐ 19.43 us ☐ 2716 dot

Sync 0.23 us 32 dot

Backp 0.57 us 80 dot

HD Start 0.00 us 0 dot

HD Width 0.00 us 0 dot

Frontp 0.34 us 48 dot

Blanking 1.14 us 160 dot

V

Input Mode ☐ msec ☒ H

Scan Progressive

Field Setting ☒ Field-1 & 2 ☐ Field-1(=2)

TV Mode Other

Frequency 29.95 Hz

Total ☐ 33.389 ms ☐ 1623 H

Disp ☐ 32.916 ms ☐ 1600 H

Sync 0.123 ms 6.0 H

Backp 0.288 ms 14 H

Serration 1H

EQP Off

EQP fp 0.000 ms 0.0 H

EQP bp 0.000 ms 0.0 H

VD Start 0.000 ms 0.0 H

VD Line 0.000 ms 0.0 H

Frontp 0.062 ms 3.0 H

Blanking 0.473 ms 23.0 H

2716x1600@59p9Hz.prg - Timing

H

Input Mode ☐ usec ☒ dot

DotClock 269.900 MHz

Repetition 1

Frequency 97.44 KHz

Period ☐ 10.26 us ☐ 2770 dot

Disp ☐ 10.06 us ☐ 2716 dot

Sync 0.07 us 20 dot

Backp 0.07 us 20 dot

HD Start 0.00 us 0 dot

HD Width 0.00 us 0 dot

Frontp 0.05 us 14 dot

Blanking 0.20 us 54 dot

V

Input Mode ☐ msec ☒ H

Scan Progressive

Field Setting ☒ Field-1 & 2 ☐ Field-1(=2)

TV Mode Other

Frequency 59.92 Hz

Total ☐ 16.688 ms ☐ 1626 H

Disp ☐ 16.421 ms ☐ 1600 H

Sync 0.051 ms 5.0 H

Backp 0.164 ms 16 H

Serration Off

EQP Off

EQP fp 0.000 ms 0.0 H

EQP bp 0.000 ms 0.0 H

VD Start 0.000 ms 0.0 H

VD Line 0.000 ms 0.0 H

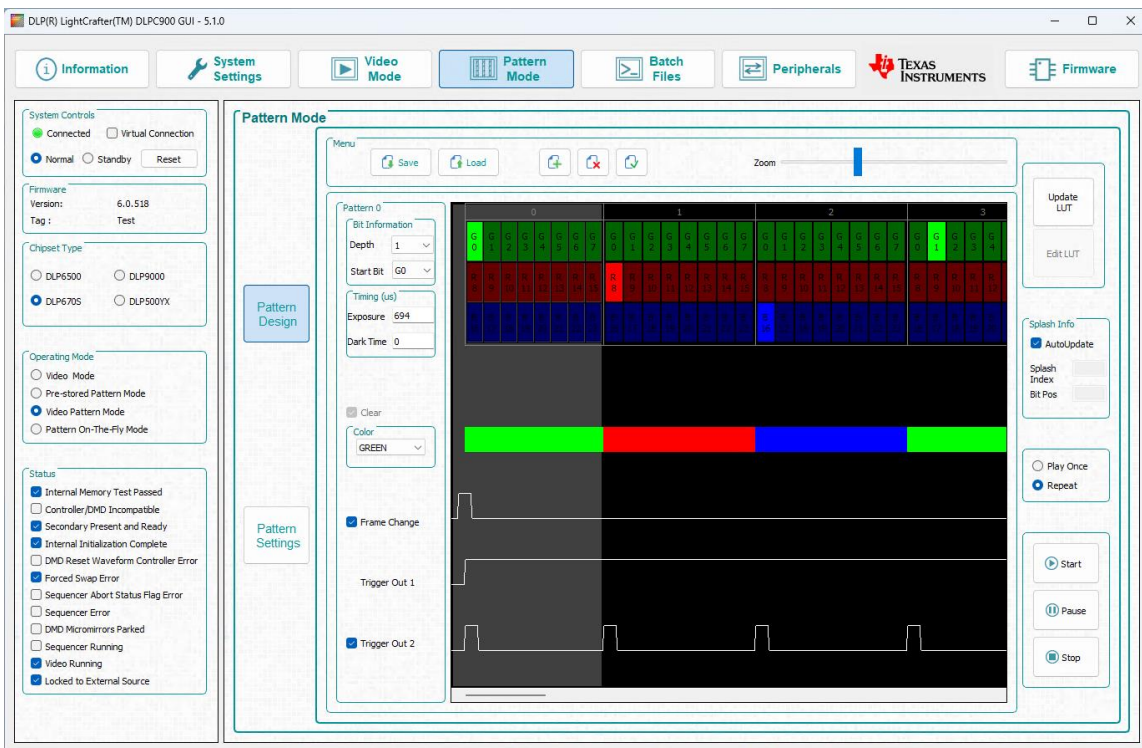
Frontp 0.051 ms 5.0 H

Blanking 0.267 ms 26.0 H

Creating Video Patterns

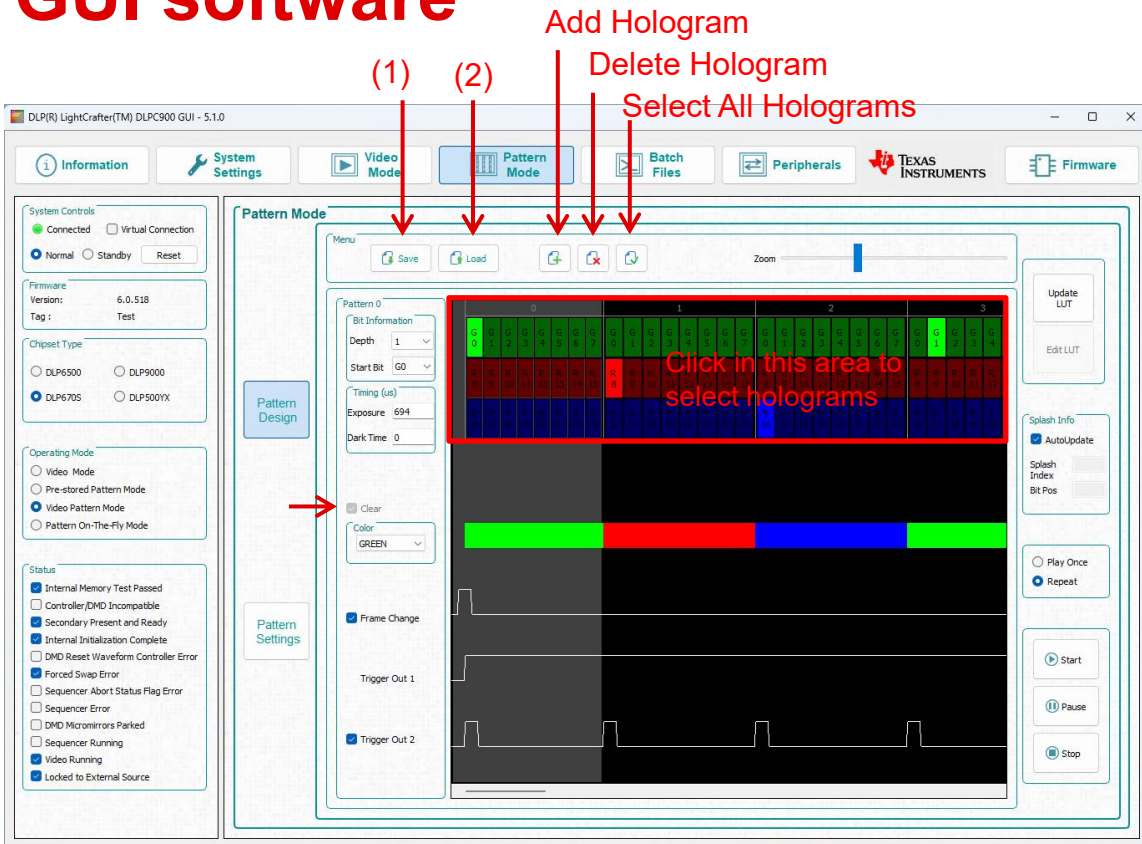
EVM Instructions – Video Pattern Mode Settings

- The GUI allows a pattern of different holograms that are input on the 24 RGB input bits to be displayed within the frame time.
- The default software loaded in the EVM has several supported base sequences (694us, 925us, etc.). This is a minimum hologram display time. All holograms displayed within the frame must be a multiple of these times and fit within the input frame time.
- Note that for hologram display (exposure) times longer than a base sequence, the EVM will reload the hologram at those intervals



EVM Instructions – GUI software

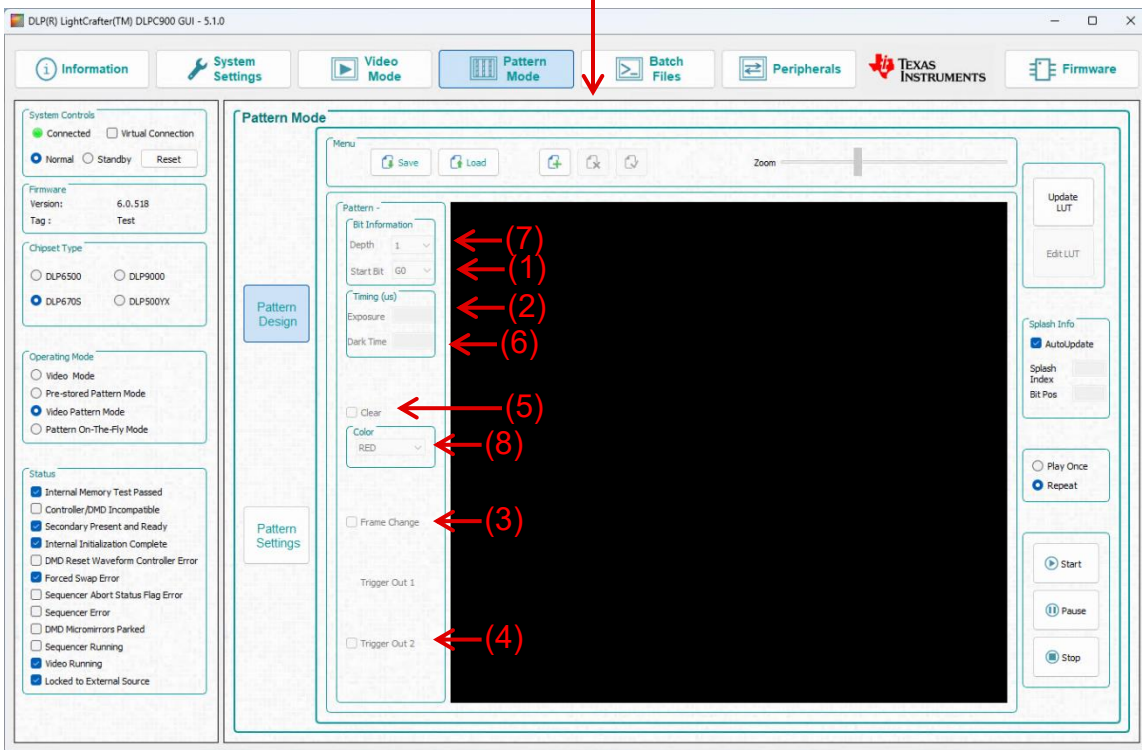
- The sequence pattern configuration can be saved (1) or restored (2) with the save and load buttons.
- Starting with a provided example pattern configuration file is recommended.
- When loading a saved configuration all the holograms will default with the clear box checked. Select all holograms and uncheck the checkbox.
- If the checkbox is grayed out add a hologram bit and then delete it. This should allow the checkboxes to be unchecked.



EVM Instructions – GUI software

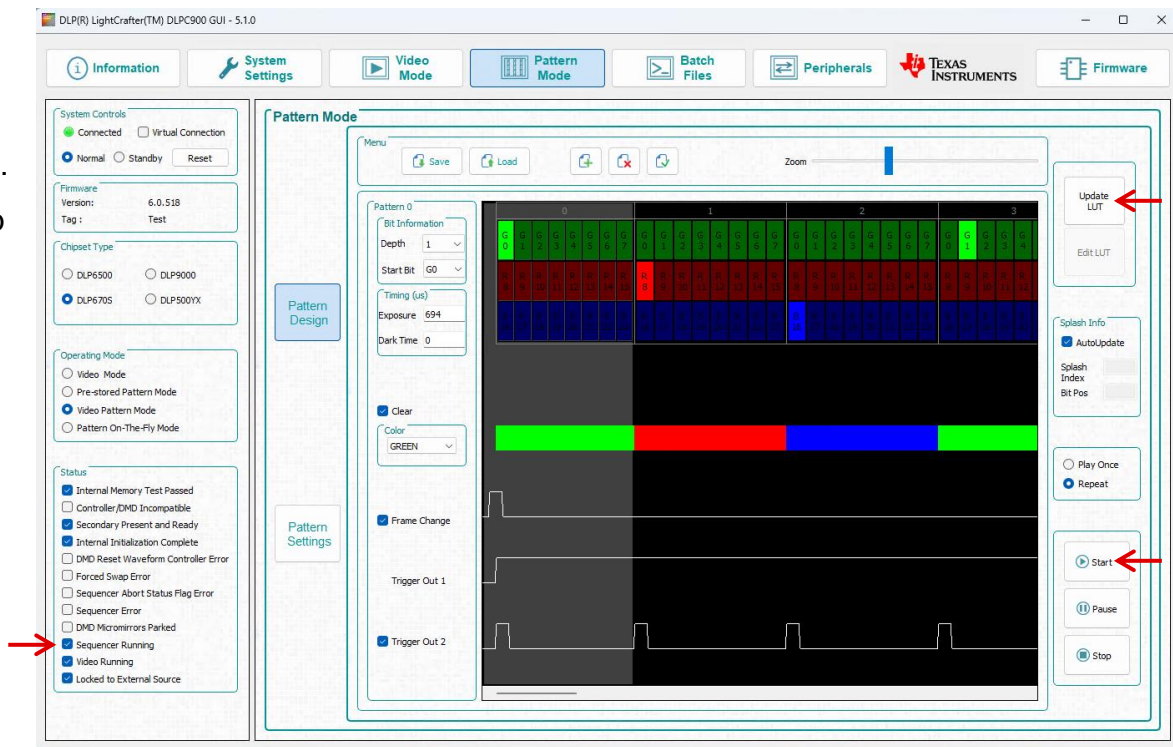
- To create a new hologram pattern click on add pattern button
- As an example, for the 694us base sequence firmware the number of holograms used should be an integer divider of 24. The holograms can be all the same or any combination of the holograms input on the 24 red, green and blue input bits (1).
- The exposure (2) for each bit must be be the same and set according to the input frame rate and the number of bits.
- The Frame Change box (3) should only be checked for the first bit of the frame.
- The Tigger Out 2 (4) is typically checked for all bits
- The clear check box (5) should be unchecked for all bits.
- For all bits set the Dark (6) time to 0 and the bit depth (7) to 1.
- Multiple bits can be selected at a time to change parameters.
- The Color (8) selection drop down only controls the which illumination enable is asserted during the hologram display

Add Hologram



EVM Instructions – Video Pattern Mode Settings

- Click on Update LUT
- Click on Start
- The Sequence Running box in the Status window should become checked.
- To change parameters click on the Stop button, change parameters, click on Update LUT and then click on Start



DLP[®] PLM Phased Reset

Achieving Continuous Signal Throughput

- Data is loaded globally during reset on TI PLM currently. This event causes all mirrors to transition to non-displaced state
- It takes $\sim 100\mu\text{s}$ to load data. Since mirrors can transition between states in $\sim 50\mu\text{s}$, mirrors reach non-displaced states
- This interrupts the outgoing signal from the PLM

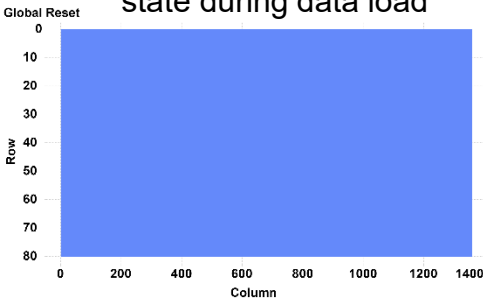
Current Operation – Mirrors return to undeflected state during data load



Achieving Continuous Signal Throughput

Current 0.67" PLM EVM

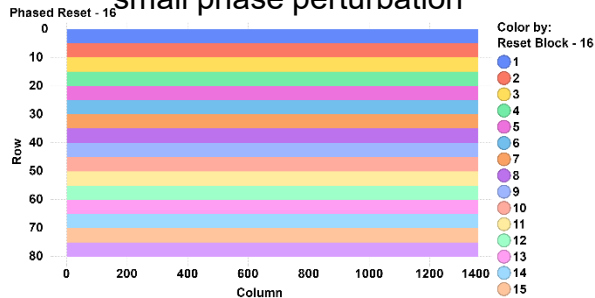
Mirrors return to undeflected state during data load



- Global reset scheme
- 1 reset block, ~90us / block
- Total ~90us data load

In Development (3Q24)

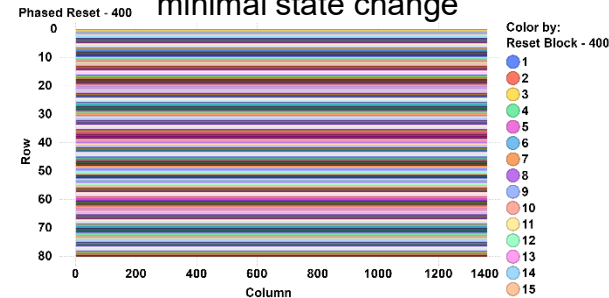
Blocks reset sequentially,
small phase perturbation



- Phased reset scheme
- 16 reset blocks, ~6us / block
- Total ~90us data load time

TBD

Blocks reset sequentially,
minimal state change



- Phased reset scheme
- *400 reset blocks, ~0.25us / block
- Total ~90us data load time

Phased reset scheme = data loaded to blocks sequentially. See 16-block details below:

- During block 1 data load, bias is removed for ~6us.
- Depending on original state, mirrors begin transition to zero-displacement state but likely do not have enough time to return to zero-displacement state before new state (bias) is applied. Some phase perturbation expected.
- New state (bias) is applied and mirrors transition to new displacement state.

Achieving Continuous Signal Throughput

**16 Block Reset – Blocks reset sequentially,
small phase perturbation**



- During block 1 data load, bias is removed for ~6 μ s
- Depending on original state, mirrors begin transition to zero-displacement state but likely do not have enough time to return to zero-displacement state before new state (bias) is applied.
- Some phase perturbation expected

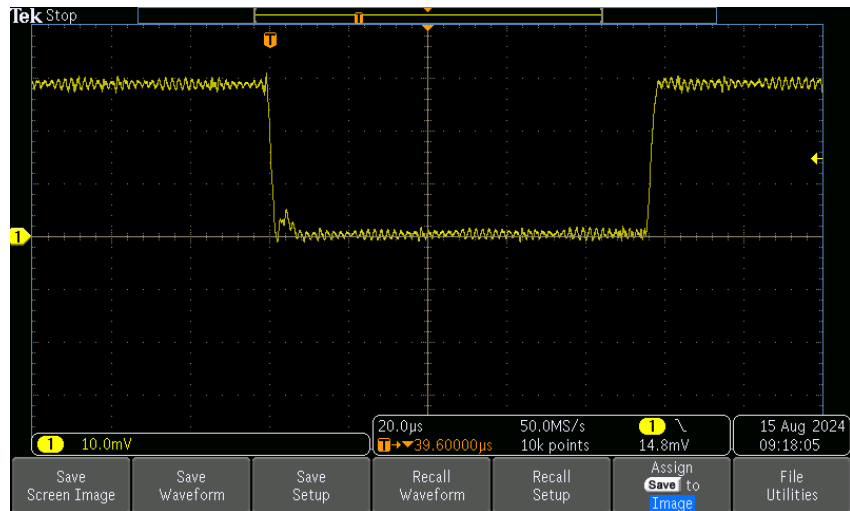
Achieving Continuous Signal Throughput

~400 Block Reset – ~0.25us / block
Blocks reset sequentially,
minimal state change/phase error

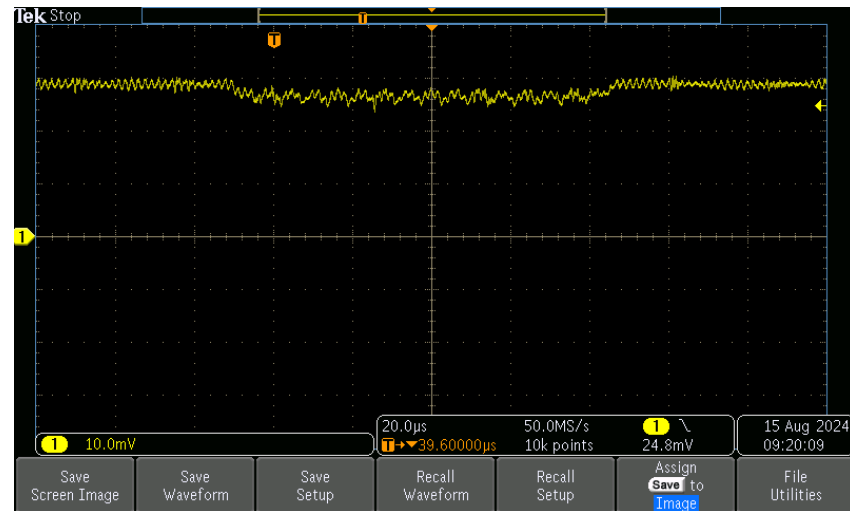


*Displacement amount is for illustration purposes

Global versus Phased Reset Waveform



Global reset waveform, 100us



Phased reset waveform, 16 resets